

An Empirical Ceiling on Star Formation Rates Recoverable from 1.4 GHz Radio Continuum Emission

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ABSTRACT

Radio continuum emission at ~ 1.4 GHz is widely used as a dust-insensitive tracer of star formation rate (SFR) and is a key science driver for forthcoming deep radio surveys. We reassess radio-based SFR inference using deep multi-wavelength data for field galaxies from VLA–COSMOS and cluster galaxies from the MeerKAT Galaxy Cluster Legacy Survey ($0 < z \lesssim 6.5$), with infrared-derived SFRs as independent references. We find that the canonical proportional ($m = 1$) radio–SFR calibration is a luminosity-dependent biased predictor of SFR when “radio-excess” sources cannot be removed prior to prediction. Including these sources recovers a best-fit linear relation that reduces bias, but retains asymmetric, luminosity-dependent scatter. We further use linear and non-linear regression models trained on radio-only inputs to quantify the recoverable variance in SFR. Across all models, performance saturates well below an idealised, scatter-limited linear mapping. As measured by R^2 (the fraction of variance explained), the gap between the model ceiling ($R^2 \lesssim 0.45$) and the idealised limit ($R^2 \simeq 0.70$) decomposes into two contributions of comparable magnitude: an irreducible noise floor set by scatter in synchrotron emissivity, and a calibration bias from the sub-unity slope. Masking identified AGN does not remove the ceiling. We additionally find that radio-excess sources are preferentially associated with early-type, passive hosts, offering a means of isolating them independently of radio properties. While radio emission remains a valuable tracer of star formation, robust SFR inference in the MeerKAT and SKA era will require multi-wavelength or probabilistic approaches accounting for intrinsic scatter and environmental modulation.

Keywords: Galaxies (573) – Galaxy Clusters (584) – Intracluster medium (858) – Star Formation (1569) – Radio Astronomy (1338)

1. INTRODUCTION

Measuring galaxy star formation rates (SFRs) across cosmic time is central to understanding how baryons cycle through galaxies and assemble stellar mass. Star formation diagnostics span ultraviolet, optical recombination lines, infrared (IR), and radio wavelengths, each sensitive to different physical timescales and systematic effects (Kennicutt 1998; Murphy et al. 2011). Among these, radio continuum emission near ~ 1.4 GHz has become particularly prominent because it is largely insensitive to dust extinction and can be observed efficiently over wide areas (e.g., Condon 1992; Murphy et al. 2011; Smolčić et al. 2017).

Radio luminosity has been used as a quantitative SFR tracer across a wide range of redshifts and environments (e.g., Seymour et al. 2008; Novak et al. 2017; Algera et al. 2020; Cochrane et al. 2023; Matthews et al. 2024), with a key requirement in that body of work being the separation of star-forming galaxies from AGN-dominated radio sources (e.g., Algera et al. 2020; Novak et al. 2017). This advantage is especially compelling in the era of deep, wide-field surveys with MeerKAT and the forthcoming Square Kilometre Array (SKA), where radio continuum will serve as a primary discovery space for star-forming galaxies across a broad range of redshifts and environments (e.g., Knowles et al. 2022).

The widespread use of radio emission as an SFR tracer is anchored in the far-infrared–radio correlation (FRC), which links dust-reprocessed emission from young massive stars to predominantly non-thermal synchrotron radiation produced by cosmic-ray electrons accelerated in supernova remnants (de Jong et al. 1985; Helou et al. 1985; Condon 1992). This empirical relation underpins commonly adopted radio–SFR calibrations that assume a monotonic, often unity-slope mapping between 1.4 GHz luminosity and star formation activity (Condon 1992; Bell 2003).

However, radio luminosity depends not only on star formation but also on magnetic field strength, cosmic-ray transport, and energy-loss processes that vary with galaxy mass, structure, redshift, and environment (Bell 2003; Lacki et al. 2010; Murphy 2009; Schleicher & Beck 2016). In dense environments, additional non-thermal emission associated with intracluster processes may further complicate interpretation (e.g., Padovani et al. 2015; Croston et al. 2019). These factors introduce intrinsic scatter and potential departures from a unity-slope relation in the radio–SFR mapping.

AGN contamination presents an additional challenge. Radio emission from accretion-driven activity can mimic or exceed star-formation-powered synchrotron emission (e.g., Delvecchio et al. 2017, 2021). A common response is to remove “radio-excess” systems, typically defined by deviations from the mean infrared–radio relation (Delhaize et al. 2017; Smolčić et al. 2017). Doing so does substantially tighten the observed radio–SFR relation, which may appear to validate the exclusion. However, radio excess alone is not a clean AGN diagnostic: it can arise from enhanced magnetic fields, efficient cosmic-ray calorimetry, and time-dependent star formation histories without any AGN contribution (Lacki & Thompson 2010; Delhaize et al. 2017). Moreover, future wide-area radio surveys may lack the necessary multi-wavelength coverage to distinguish AGN-driven emission from that of star-forming galaxies.

Beyond sample selection, a more fundamental issue concerns the informational content of radio luminosity itself. Much of the literature has treated the radio–SFR relation primarily as a calibration problem, measuring slopes and scatter under the assumption of a deterministic mapping (Condon 1992; Bell 2003; Delhaize et al. 2017; Novak et al. 2017). Here we instead pose an information-content question: *how much of the variance in SFR is recoverable from radio continuum observables alone?* We directly quantify this recoverable variance using deep multi-wavelength data from the VLA–COSMOS field and the MeerKAT Galaxy Cluster Legacy Survey, extending to $z \lesssim 6.5$.

The structure of this paper is as follows. In Section 2, we describe the construction of the field and cluster samples, the derivation of infrared-based SFRs, and our treatment of AGN and radio-excess sources. In Section 3, we outline the machine learning framework, define the concept of recoverable variance, and detail the training and evaluation strategy used to quantify the informational content of radio luminosity. Section 4 presents the empirical radio–SFR relation, the perfect-mapping analysis, the predictive performance of radio-only models, and tests of environmental dependence in cluster environments. In Section 5, we interpret the observed variance ceiling in terms of cosmic-ray physics, magnetic field variation, star formation history, and environmental effects, and discuss implications for current and forthcoming radio surveys. Finally, Section 6 summarizes our principal findings and outlines consequences for radio-based SFR inference in the MeerKAT and SKA era. Throughout this work, we assume a flat Λ CDM cosmology with parameters $H_0 = 67.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_M = 0.307$, and $\Omega_\Lambda = 0.693$, consistent with *Planck* 2018 results (Planck Collaboration et al. 2020).

2. DATA

Our analysis combines deep radio continuum observations with independent infrared-based star formation rate (SFR) measurements across a wide range of environments and redshifts. We draw on two complementary datasets: the VLA–COSMOS 3 GHz Large Project, which provides a well-characterized field sample with extensive multi-wavelength coverage, and the MeerKAT Galaxy Cluster Legacy Survey (MGCLS), which enables a direct assessment of environmental effects in dense cluster regions. Together, these datasets allow us to evaluate the reliability of radio luminosity as a star formation tracer under both ideal and realistic observational conditions.

3.

A central requirement of this work is that SFR estimates be derived independently of the radio data used for prediction. In the COSMOS field, this is achieved via full infrared SED fitting incorporating far-infrared data and explicit AGN components, yielding robust, dust-unbiased SFRs (Delvecchio et al. 2017; Berta et al. 2013). In the cluster sample, where homogeneous far-infrared coverage is unavailable, we adopt mid-infrared SFR estimates based on *WISE* photometry and empirical calibrations (Cluver et al. 2017). Although these two approaches differ in methodology, both provide physically motivated benchmarks against which the information content of radio luminosity can be tested.

Radio luminosities throughout this work are quoted as $\log_{10}(L_{1.4\text{ GHz}} [\text{W Hz}^{-1}])$. Where flux densities are required for comparison with observed catalogues, these are given in units of mJy and are clearly distinguished from luminosities in the text and figures. Radio luminosities are computed assuming a power-law radio spectrum, $S_\nu \propto \nu^\alpha$, with spectral index $\alpha = -0.7$, typical of synchrotron emission from star-forming galaxies (Condon 1992). While individual sources may deviate from this assumption, we show that our qualitative conclusions are insensitive to reasonable variations in spectral index and redshift uncertainties.

In the following subsections, we describe the construction of the COSMOS and cluster samples, the environmental characterisation of cluster galaxies (including cluster properties and source counts), the derivation of infrared-based SFRs for the cluster sample, and our treatment of AGN and radio-excess sources.

2.1. VLA–COSMOS 3 GHz Large Project

The VLA–COSMOS 3 GHz Large Project provides deep radio continuum imaging over 2 deg^2 with a median rms sensitivity of $\sim 2.3\text{ }\mu\text{Jy beam}^{-1}$ and an angular resolution of $0''.75$ (Smolčić et al. 2017). The survey detects $\sim 10^4$ radio sources above 5σ significance, spanning star-forming galaxies and AGN across a broad redshift range.

We select galaxies with reliable 3 GHz detections above 5σ (corresponding to a flux density limit of $\sim 11.5\text{ }\mu\text{Jy}$ at the median rms), measured 1.4 GHz flux densities, redshift estimates, and infrared-derived SFRs. Multi-wavelength counterparts are obtained via likelihood-ratio matching (Sutherland & Saunders 1992) to the COSMOS2015 catalogue (Laigle et al. 2016), providing photometric redshifts, stellar masses, and ancillary data. This sample forms the basis for training and validation of our machine learning models.

Infrared luminosities and SFRs are taken from Delvecchio et al. (2017), who derived total infrared (8–1000 μm) luminosities using the SED3FIT SED-fitting framework (Berta et al. 2013). This approach simultaneously models stellar emission, dust-reprocessed star formation, and AGN components using *Spitzer* and *Herschel* photometry, ensuring that the inferred SFRs are not contaminated by AGN-heated dust (Fritz et al. 2006; Feltre et al. 2012). These IR-based SFRs provide a robust, independent reference against which radio-based predictions can be evaluated.

The full COSMOS sample used for model training and evaluation comprises $N = 7,729$ sources spanning $0 < z \lesssim 6.5$. Models are trained on a randomly selected 80% subset ($N = 6,183$) and evaluated on the held-out 20% test set ($N = 1,546$). The test set spans $z = 0.10\text{--}5.37$ (median $z = 1.24$), with 71% photometric and 29% spectroscopic redshifts, and breaks down as 1,128 star-forming galaxies (73%), 170 radio-excess non-AGN (11%), and 248 AGN (16%), identified via X-ray (128), mid-infrared (55), and SED-based (177) diagnostics consistent with Delvecchio et al. (2017). All figures and R^2 values are evaluated on this held-out test set unless otherwise stated; the scatter parameters quoted for the perfect-mapping analysis (Section 4.2) are derived from this same subset.

2.2. MeerKAT Galaxy Cluster Legacy Survey

To assess environmental effects, we apply our analysis to 24 galaxy clusters drawn from the MeerKAT Galaxy Cluster Legacy Survey (Knowles et al. 2022). MGCLS provides 1.28 GHz radio continuum imaging of 115 clusters over $> 20\text{ deg}^2$, with typical rms sensitivities of $2\text{--}5\text{ }\mu\text{Jy beam}^{-1}$ and angular resolution of $\sim 8''$.

Compact radio sources are identified using PYBDSF (Mohan & Rafferty 2015). Optical and infrared counterparts are obtained via likelihood-ratio matching (Sutherland & Saunders 1992) to DES (Dark Energy Survey Collaboration et al. 2016) and *WISE* (WISE Team 2020) catalogues. The resulting radio-selected samples include star-forming galaxies, AGN, and systems affected by cluster-specific processes such as ram-pressure stripping, shocks, and diffuse synchrotron emission (e.g., Dressler 1980; Cuciti et al. 2021).

MGCLS and COSMOS-VLA enable a direct comparison of the radio–SFR relation inside and outside cluster environments, where additional non-star-forming radio emission from relativistic electrons in the intracluster medium may further complicate the interpretation of radio luminosity (e.g., Croston et al. 2019).

2.3. Cluster Properties, Environmental Characterisation, and Source Counts

For each cluster, we adopt published redshifts and characteristic radii (R_{500}) from X-ray, lensing, and dynamical analyses (e.g., Piffaretti et al. 2011; Cuciti et al. 2021; Mantz et al. 2016; Umetsu et al. 2022). Here, R_{500} denotes the radius within which the mean enclosed mass density is 500 times the critical density of the Universe at the cluster redshift, and is commonly used as a physically motivated scale that approximates the virialized region accessible to X-ray observations. Where necessary, we approximate $R_{500} \simeq R_{200}/1.5$ ($R_{500}/R_{200} \approx 0.65$), consistent with Navarro, Frenk, & White (NFW; 1997) halo profiles and typical cluster concentrations of $c_{200} \sim 4\text{--}5$. Projected cluster-centric

distances are computed for all radio sources, allowing the sample to be divided into galaxies located within R_{500} (cluster cores) and those at larger radii. This separation enables an explicit test of whether dense environments degrade the reliability of radio luminosity as a tracer of star formation. To characterize cluster environments, we incorporated published R_{500} and mass parameters from:

- The *Meta-catalogue of X-ray Detected Clusters of Galaxies (MCXC)* (Piffaretti et al. 2011);
- Mass-selected cluster samples and radio halo studies (Cuciti et al. 2021);
- Detailed analyses of individual systems (e.g., Abell 133, Abell 194, Abell 209, El Gordo; Fujita et al. 2013; Thuruthipilly & Junais 2024; Cuciti et al. 2021; Zitrin et al. 2013);
- Environmental studies of cluster galaxy evolution (Venkatapathy et al. 2017);
- Additional dynamical, lensing and X-ray reconstructions (Mantz et al. 2016; Urdampilleta et al. 2021; Umetsu et al. 2022).

Table 1 summarizes the MGCLS cluster fields used in this work, including pointing coordinates, redshifts, adopted R_{500} values, and the corresponding literature references.

Table 1. Galaxy Cluster Properties

Cluster	RA (deg)	Dec (deg)	Redshift	R_{500} (Mpc)	No. of Sources	R_{500} Ref.
Abell 22	5.1608	-25.7220	0.142	1.0560	3417	Arnaud et al. (2010)
Abell 85	10.4529	-9.3180	0.055	1.2000	2470	Venkatapathy et al. (2017)
Abell 133	15.6879	-21.8800	0.057	1.0020	2930	Fujita et al. (2013)
Abell 194	21.4458	-1.3731	0.018	0.6549	907	Thuruthipilly & Junais (2024)
Abell 209	22.9896	-13.5764	0.210	0.8867	2481	Cuciti et al. (2021)
Abell 2485	342.1371	-16.1062	0.247	1.0622	3627	Piffaretti et al. (2011)
Abell 2597	351.3321	-12.1244	0.085	1.1066	1011	Piffaretti et al. (2011)
Abell 2645	355.3200	-2.0975	0.251	1.0897	2009	Piffaretti et al. (2011)
Abell 2667	357.9196	-26.0836	0.230	1.3031	4001	Piffaretti et al. (2011)
Abell 2744	3.5671	-30.3830	0.308	1.2359	3048	Piffaretti et al. (2011)
Abell 2751	4.0580	-31.3885	0.107	0.7431	2111	Piffaretti et al. (2011)
Abell 2811	10.5368	-28.5358	0.108	1.0355	773	Piffaretti et al. (2011)
Abell 2895	19.6454	-26.9731	0.227	1.0896	3383	Piffaretti et al. (2011)
Abell 3365	87.0500	-21.9350	0.093	0.8100	1067	Urdampilleta et al. (2021)
Abell 3376	90.4256	-39.9851	0.046	0.8726	2421	Piffaretti et al. (2011)
Abell 3558	201.9783	-31.4922	0.048	1.1010	2645	Piffaretti et al. (2011)
Abell 3562	202.7833	-31.6713	0.049	0.9265	2547	Piffaretti et al. (2011)
Abell 3667	303.1403	-56.8406	0.056	1.1990	1176	Piffaretti et al. (2011)
Abell 370	39.9604	-1.5856	0.375	1.0933	1407	Umetsu et al. (2022)
Abell 4038	356.8796	-28.2028	0.028	0.8863	3593	Piffaretti et al. (2011)
Abell 521	73.5358	-10.2442	0.253	1.2400	2423	Cuciti et al. (2021)
Abell 545	86.7571	-25.6164	0.154	1.3500	2548	Mantz et al. (2016)
El Gordo	15.7188	-49.2495	0.870	1.1500	3127	Zitrin et al. (2013)
J0014.3-6604	3.5767	-66.0775	0.155	1.0117	4459	Piffaretti et al. (2011)

Notes: (1) Cluster names listed alphabetically and by ascending Abell number, with non-Abell clusters listed last; (2) and (3) MeerKAT pointing coordinates: J2000 RA and Dec (Knowles et al. 2022); (4) Cluster redshift (Knowles et al. 2022); (5) R_{500} ; (6) Number of sources in the cluster field; (7) R_{500} reference.

For completeness, Table 2 summarizes the number of compact radio sources detected in each MGCLS cluster field, the number of those sources with reliable *WISE* infrared counterparts, and the final matched sample used in the machine-learning analysis. Infrared counterparts are identified only at the positions of radio-detected sources; consequently,

Table 2. Source counts per cluster. Columns list the total number of radio sources identified in the MGCLS compact source catalogue (N_{radio}), the number of radio sources with reliable WISE infrared detections (N_{WISE}), and the final matched sample used in the machine-learning analysis ($N_{\text{radio+WISE}}$).

Cluster	N_{radio}	N_{WISE}	$N_{\text{radio+WISE}}$
Abell 133	5040	2930	2930
Abell 194	1422	907	907
Abell 209	4456	2481	2481
Abell 22	6119	3417	3417
Abell 2485	7170	3627	3627
Abell 2597	1661	1011	1011
Abell 2645	3488	2009	2009
Abell 2667	7307	4001	4001
Abell 2744	5224	3048	3048
Abell 2751	3610	2111	2111
Abell 2811	1256	773	773
Abell 2895	6059	3383	3383
Abell 3365	2050	1067	1067
Abell 3376	4370	2422	2422
Abell 3558	5644	2645	2645
Abell 3562	5442	2547	2547
Abell 3667	2099	1176	1176
Abell 370	2261	1405	1405
Abell 4038	6197	3593	3593
Abell 521	4342	2423	2423
Abell 545	5039	2548	2548
Abell 85	4629	2470	2470
El Gordo	5558	3127	3127
J0014.3–6604	8456	4459	4459

All WISE sources are associated with radio detections by construction; infrared counterparts and star formation rates were derived only for radio-selected sources. Consequently, $N_{\text{WISE}} = N_{\text{radio+WISE}}$ for all clusters.

all infrared sources considered in this work are associated with radio detections by construction. The radio+WISE sample therefore represents a subset of the radio-selected population, rather than an independent parent sample. These source counts are provided to document the statistical weight of each cluster field and to facilitate reproducibility of the analysis.

2.4. Infrared-based Star Formation Rates

For the cluster sample, star formation rates are estimated using *WISE* *W4* ($22\mu\text{m}$) photometry and the empirical calibration of Cluver et al. (2017). Observed *W4* magnitudes are converted to flux densities using the *WISE* zero-point of $F_{0,W4} = 8.283\text{ Jy}$ (Wright et al. 2010). The flux density is then converted to a monochromatic luminosity using the luminosity distance at the cluster redshift.

To account for bandpass shifting with redshift, we apply mid-infrared *k*-corrections derived from fits to the observed *W1*–*W4* photometry using a simple two-component SED model, consisting of a Rayleigh–Jeans stellar component and a power-law dust component. The best-fitting SED is evaluated at an observed wavelength of $22(1+z)\mu\text{m}$, corresponding to rest-frame $22\mu\text{m}$ emission. While monochromatic mid-infrared estimators are sensitive to dust temperature, metallicity, and star formation timescales, they provide a practical and widely used SFR proxy in the absence of homogeneous far-infrared coverage (Cluver et al. 2017).

Although COSMOS and cluster SFRs are derived using different methodologies, these differences primarily introduce additional scatter and modest systematic offsets rather than order-of-magnitude discrepancies. Since our analysis

focuses on correlations, variance, and predictive limits rather than absolute SFR normalization, these differences do not qualitatively affect our conclusions.

To convert luminosities to flux densities for direct comparison with observed catalogues, we adopt a flat Λ CDM cosmology with parameters from the *Planck* Collaboration (Planck Collaboration et al. 2020) results: $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_m = 0.315$. For each source at redshift z , we compute the luminosity distance $D_L(z)$ using *Astropy*’s cosmology utilities (Astropy Collaboration et al. 2013, 2018).

2.5. Infrared–Radio Ratio and Radio-excess Sources

The relative strength of infrared and radio emission is commonly quantified using the infrared–radio ratio q (Helou et al. 1985), defined as

$$q_{\text{IR}} \equiv \log_{10} \left(\frac{L_{\text{IR}}}{3.75 \times 10^{12} \text{ W}} \right) - \log_{10} \left(\frac{L_{1.4 \text{ GHz}}}{\text{W Hz}^{-1}} \right), \quad (1)$$

where L_{IR} is the total infrared luminosity integrated over 8–1000 μm and $L_{1.4 \text{ GHz}}$ is the monochromatic radio luminosity at 1.4 GHz in units of W Hz^{-1} .

Star-forming galaxies exhibit a characteristic distribution in q_{IR} with intrinsic scatter of ~ 0.2 – 0.3 dex (Yun et al. 2001; Bell 2003). Systems with lower q_{IR} values are often labelled as radio-excess sources and frequently interpreted as AGN-dominated (Delvecchio et al. 2021; Smolčić et al. 2017).

However, radio excess does not uniquely imply AGN activity. Intrinsic scatter in the IR–radio correlation, compact starbursts with enhanced magnetic fields, efficient cosmic-ray calorimetry, and secondary production of relativistic electrons and positrons in dense gas can all increase synchrotron emissivity per unit SFR (e.g., Thompson et al. 2007; Murphy 2009; Yoast-Hull et al. 2016). In addition, time-dependent star formation histories can temporarily depress infrared emission relative to radio emission, producing offsets of up to $\lesssim 0.5$ dex below the mean IR–radio relation without any AGN contribution (Bressan et al. 2002; Delhaize et al. 2017). Only $\sim 0.7\%$ of the radio-excess sources in our sample lie beyond 0.5 dex of the non-AGN mean.

We find that removing radio-excess sources does substantially tighten the radio–SFR relation (Table 4). Applying the standard 2σ threshold ($q_{\text{IR}} < \bar{q} - 2\sigma_q$; Delvecchio et al. 2021; Smolčić et al. 2017) removes 378 sources (4.9% of the sample) and raises the Pearson correlation from $r = 0.652$ to $r = 0.789$, the best-fit slope from $m = 0.574$ to $m = 0.724$, and the best-performing model R^2 from 0.459 to 0.633. A stricter 1σ cut (removing 14% of the sample) raises these further to $r = 0.876$, $m = 0.811$, and $R^2 = 0.739$. The relation does tighten, and tightens substantially. We nevertheless retain radio-excess sources in our primary analysis for two reasons, which together constitute a physical and methodological argument against their routine exclusion.

First, the improvement is an artefact of what radio-excess sources *are*, not evidence of what they *are not*. Radio-excess is defined as elevated radio emission relative to infrared emission, that is, these sources lie off the radio–SFR relation by construction. Removing sources that are offset from a relation will always tighten that relation, regardless of why they are offset. The standard literature practice of applying a q_{IR} cut (e.g., Delvecchio et al. 2017; Smolčić et al. 2017; Delhaize et al. 2017; Novak et al. 2017) therefore does not demonstrate that the resulting sample is physically clean; it demonstrates that the outliers have been removed. The degree of tightening is proportional to how far the excluded sources lie from the relation, not to the physical validity of the exclusion criterion (Condon 1992).

Second, and more decisively, the physical nature of these sources does not support their blanket classification as AGN. We characterise their morphology not as a stand-alone structural study but as targeted evidence for the single question at issue here, namely whether radio-excess sources are a physically distinct contaminant population that justifies their removal from radio-based SFR inference. To characterise the morphological properties of radio-excess hosts, we use publicly available *Hubble Space Telescope* (HST) ACS F814W imaging from the COSMOS survey (Scoville et al. 2007), which provides $\sim 0''.09$ resolution I-band photometry across the full VLA–COSMOS footprint.

Non-parametric morphological statistics, specifically the Gini coefficient (G ; Lotz et al. 2004), concentration index (C ; Conselice 2003), and rotational asymmetry (A ; Conselice 2003), are computed for each source using the STATMORPH package (Rodriguez-Gomez et al. 2019), which performs pixel-level measurements on background-subtracted, PSF-corrected cutouts. All measurements use a signal-to-noise threshold of $S/N \geq 2.5$ per pixel and a Petrosian elliptical aperture to define the galaxy extent. As an independent check on the non-parametric statistics, we cross-match the radio-excess and control samples against the Zurich Estimator of Structural Types (ZEST) catalogue (Scarlata et al. 2007), which provides visual and PCA-based morphological classifications (early-type, disc, and irregular) for $\sim 56\,000$ COSMOS galaxies from the same HST/ACS F814W imaging. The ZEST early-type fraction (Type 1) among

radio-excess hosts is significantly higher than among matched controls, providing independent corroboration that these sources preferentially occupy morphologically undisturbed, bulge-dominated hosts rather than the disturbed or disc-dominated systems typical of merger-triggered AGN.

To isolate the effect of radio excess from confounding variables, each radio-excess source ($N_{\text{rx}} = 1003$) is matched one-to-one without replacement to a unique non-radio-excess control in standardised ($\log M_*$, z , $\log L_{\text{TIR}}$) space using nearest-neighbour matching with a maximum tolerance of 0.1σ per dimension. This ensures that any morphological difference between the two populations cannot be attributed to differences in stellar mass, redshift, or infrared luminosity. We additionally separate the radio-excess population by spectral index α_{radio} , computed between the 1.4 and 3 GHz flux densities, defining flat-spectrum sources as those with $\alpha_{\text{radio}} \geq -0.5$ ($N_{\text{rx}} = 139$) and steep-spectrum sources as those with $\alpha_{\text{radio}} < -0.5$ ($N_{\text{rx}} = 864$).

Our morphological characterisation (Table 3) shows that, after this matching, radio-excess sources show significantly higher Gini coefficients ($d = +0.22$; 95% CI $[+0.13, +0.31]$) and concentration indices ($d = +0.18$; $[+0.10, +0.27]$) than matched non-radio-excess controls, and are simultaneously smoother ($d_A = -0.14$; $[-0.23, -0.05]$). For the flat-spectrum sub-sample, the sources most likely to harbour compact, self-absorbed radio emission, the asymmetry suppression is extreme ($d = -0.87$; 95% CI $[-1.12, -0.64]$; $p = 8 \times 10^{-13}$): their median asymmetry is barely half that of individually matched controls at fixed mass, redshift, and SFR. These results are consistent with the ZEST classification of radio-excess hosts as preferentially early-type (Scarlata et al. 2007). These are not the morphologically disturbed, merger-triggered AGN hosts that dominate the radio-loud AGN literature (Holt et al. 2008; Hardcastle & Croston 2020; O’Dea & Saikia 2021). They are compact, smooth, and centrally concentrated, morphological properties consistent with passive early-type galaxies or systems with low-luminosity AGN sustained by secular gas supply rather than major mergers (Best et al. 2005; Croton et al. 2006; Hardcastle & Croston 2020). In the absence of resolved morphological or spectroscopic classification for all sources, classifying them as AGN contaminants and discarding them is not warranted. Retaining them allows us to evaluate the reliability of radio luminosity as a star formation tracer under the realistic and unavoidable conditions present in wide-field radio surveys.

Table 3. Morphological properties of radio-excess hosts relative to matched non-radio-excess controls, measured from HST/ACS F814W imaging (Scoville et al. 2007) using STATMORPH (Rodriguez-Gomez et al. 2019), with independent corroboration from the ZEST morphological catalogue (Scarlata et al. 2007). Each radio-excess source is matched one-to-one to a unique non-radio-excess control in standardised ($\log M_*$, z , $\log L_{\text{TIR}}$) space. Cohen’s d quantifies the standardised mean difference (positive = radio-excess hosts have larger values); 95% confidence intervals are from 5000 bootstrap resamples. KS p -values test the null hypothesis that the two distributions are drawn from the same parent population. All three effects are statistically robust and well-separated from zero. The flat-spectrum sub-sample ($\alpha_{\text{radio}} \geq -0.5$; $N_{\text{rx}} = 139$) shows extreme asymmetry suppression, consistent with morphologically undisturbed early-type hosts.

Sub-sample	Morphology metric	Cohen’s d	95% CI	KS p
Full matched sample ($N_{\text{rx}} = 1003$)	Gini coefficient	+0.22	$[+0.13, +0.31]$	4.6×10^{-6}
	Concentration	+0.18	$[+0.10, +0.27]$	6.2×10^{-4}
	Asymmetry	−0.14	$[-0.23, -0.05]$	3.6×10^{-4}
Steep-spectrum sub-sample ($\alpha_{\text{radio}} < -0.5$; $N_{\text{rx}} = 864$)	Gini coefficient	+0.19	$[+0.10, +0.29]$	3.1×10^{-4}
	Concentration	+0.16	$[+0.07, +0.26]$	1.2×10^{-3}
	Asymmetry	−0.12	$[-0.22, -0.03]$	1.1×10^{-2}
Flat-spectrum sub-sample ($\alpha_{\text{radio}} \geq -0.5$; $N_{\text{rx}} = 139$)	Gini coefficient	+0.28	$[+0.05, +0.53]$	1.5×10^{-2}
	Concentration	+0.35	$[+0.11, +0.60]$	1.0×10^{-2}
	Asymmetry	−0.87	$[-1.12, -0.64]$	8.0×10^{-13}

Notes: Positive d indicates radio-excess hosts are more compact or concentrated; negative d indicates they are smoother. The flat-spectrum asymmetry suppression ($d = -0.87$) is the dominant structural signal: the median asymmetry of flat-spectrum radio-excess hosts is barely half that of their individually matched controls, robust to spectral-index significance thresholds up to $2\sigma_\alpha$ ($d = -0.57$, $p = 0.003$). These morphological properties are consistent with passive early-type galaxies or systems with low-luminosity AGN fuelled through secular processes, rather than merger-triggered AGN or compact starbursts (Best et al. 2005; Croton et al. 2006; van der Wel et al. 2014; Hardcastle & Croston 2020), and are corroborated by the higher ZEST early-type fraction (Type 1; Scarlata et al. 2007) among radio-excess hosts relative to matched controls.

Table 4. Effect of radio-excess source removal on the radio–SFR relation and predictive performance. Radio-excess sources are defined as $q_{\text{IR}} < \bar{q} - N\sigma_q$ where $\bar{q} = 2.076$ and $\sigma_q = 0.606$ for the VLA–COSMOS sample. R^2 values are test-set scores in $\log_{10}(\text{SFR})$ space. CI half-widths are means across luminosity bins (see Table 8). The improvement in r , slope, and R^2 upon removal reflects the fact that radio-excess sources are offset from the radio–SFR relation by definition; removing outliers necessarily tightens any relation regardless of the physical cause of the offset. The retained AGN (identified via X-ray, MIR, and SED criteria) do not drive the improvement, confirming that the radio-excess and AGN populations are not equivalent.

	Full sample	Non-excess (2σ)	Non-excess (1σ)
N (sources)	7729	7351	6650
Fraction of full sample	100%	95.1%	86.0%
Best-fit slope m	0.574	0.724	0.811
Pearson r	0.652	0.789	0.876
Mean 68% CI half-width (dex)	0.123	0.106	0.095
Mean 95% CI half-width (dex)	0.420	0.365	0.297
<i>Test-set R^2 by model</i>			
Random Forest	0.198	0.408	0.584
Linear Regression	0.332	0.608	0.730
Support Vector Regression	0.459	0.633	0.739
k -Nearest Neighbors	0.355	0.510	0.663
Gradient Boosting	0.446	0.604	0.725
Neural Network (MLP)	0.438	0.616	0.733

Notes: The 2σ cut corresponds to $q_{\text{IR}} < 0.863$; the 1σ cut to $q_{\text{IR}} < 1.469$. The improvement in R^2 upon removal (e.g. from 0.459 to 0.633 for SVR) reflects the exclusion of sources that are offset from the mean relation, not a demonstration that those sources are AGN. Even after removal, AGN identified independently via X-ray and MIR criteria remain in the sample, yet the relation tightens substantially: the radio-excess and AGN populations are therefore not the same population.

2.6. AGN Identification

AGN in the COSMOS sample are identified following the multi-wavelength criteria described by Smolčić et al. (2017). X-ray AGN are selected using an intrinsic luminosity threshold of $L_X \geq 10^{42} \text{ erg s}^{-1}$ in the 0.5–8 keV band (Szokoly et al. 2004), a regime where X-ray emission from star formation is negligible (Symeonidis et al. 2014). Additional AGN are identified via mid-infrared colour diagnostics (Donley et al. 2012) and SED fitting that explicitly includes an AGN component (Delvecchio et al. 2014). We caution, however, that even the exceptionally deep, uniform multi-wavelength coverage of COSMOS does not guarantee a complete AGN census. Low-luminosity, heavily obscured, or radiatively inefficient AGN can evade X-ray, mid-infrared, and SED selection simultaneously (e.g., Hickox & Alexander 2018; Padovani et al. 2017).

We are deliberately conservative about what our result claims in light of this. We do *not* assert that the variance ceiling is purely a consequence of star-formation physics; what we establish is the weaker and more robust statement that the ceiling persists *after* removing every independently identifiable AGN, using the most complete classification available in any deep field. A residual population of undetected AGN may contribute to the radio emission of nominally star-forming sources, and to the floor we measure. Rather than undermining our conclusion, this possibility reinforces its central implication for survey science: if a residual contaminant survives even the COSMOS census, then in the wide, shallow fields that will dominate MeerKAT and SKA SFR catalogues, where comparable ancillary data do not exist, the contamination can only be larger and the recoverable-variance ceiling correspondingly lower. Radio-only SFR inference is therefore limited whether the residual scatter is astrophysical or contaminant in origin, and a survey cannot tell the two apart from radio data alone, which is precisely the operational problem we highlight.

We do not attempt to identify radio-loud AGN in the cluster sample. Uniform deep X-ray and far-infrared data required for robust AGN classification are unavailable across all cluster fields, and radio luminosity alone does not provide an unambiguous diagnostic of AGN activity (Padovani et al. 2015; Hardcastle et al. 2025). While extremely radio-luminous sources are likely AGN-dominated, defining a physically motivated and redshift-independent luminosity threshold is non-trivial, particularly given the wide range of galaxy masses, environments, and potential contributions from star-formation-driven processes. We note that the brightest cluster galaxy (BCG) is frequently radio-loud (Best et al. 2007; Hogan et al. 2015), and that excluding the BCG in each field would be a natural first step toward reducing

AGN contamination at the high-luminosity end; our cluster analysis retains all radio-selected sources and therefore provides a conservative upper bound on this contamination.

AGN contamination is, moreover, expected to be *more* severe in clusters than in the COSMOS field. Dense environments preferentially host radio AGN (Best et al. 2005; Croston et al. 2019), and the sparse cluster coverage of the COSMOS field means that our COSMOS-derived AGN census provides only limited guidance on the level of contamination present in the MGCLS cluster cores. Removing AGN from the cluster sample, were uniform classification possible, would therefore plausibly be more fruitful than in the field. Because such classification is not available, we apply our models to the cluster sample without AGN removal, which provides a realistic, if conservative, characterisation of what radio-based SFR inference can achieve under typical wide-field survey conditions where AGN cannot be cleanly excised.

Crucially, the

inability to classify cluster AGN does not undermine our central result. In the COSMOS field, where robust multi-wavelength AGN identification is available, masking confirmed AGN does not recover a tight radio–SFR relation: predictive performance remains limited at $R^2 \lesssim 0.45$ even after AGN removal (Section 4.3). The variance ceiling in COSMOS is therefore not driven by *identifiable* AGN contamination. The radio–SFR relation is consequently not solely a problem of obvious AGN contamination, but reflects a broader physical decoupling between radio luminosity and star formation (Lacki et al. 2010; Murphy 2009), plausibly compounded by undetected AGN whose prevalence is expected to grow in the shallower, less uniformly characterised fields that will dominate future radio surveys.

3. METHODS

The aim of this work is to quantify the maximum amount of information about star formation that can be extracted from radio continuum emission alone. To address this, we adopt a machine learning (ML) framework that imposes minimal assumptions on the functional form of the radio–SFR relation. If radio luminosity encodes sufficient information about star formation, flexible non-linear models should be able to recover it. Conversely, systematic failure across a broad class of models indicates that the limitation is physical rather than methodological. All models are trained using the VLA–COSMOS field sample, where high-quality infrared-derived SFRs and redshift information are available, and are evaluated on both held-out COSMOS data and independent cluster samples from the MGCLS. Throughout, infrared-based SFRs are treated as reference quantities, and radio luminosities as the predictors whose informational content is being tested.

3.1. Input Features and Target Variable

The target variable for all supervised learning experiments is the base-10 logarithm of the star formation rate, $\log_{10}(\text{SFR}/\text{M}_{\odot} \text{ yr}^{-1})$, where taking the logarithm reduces the dynamic range of the data and mitigates heteroscedastic scatter (Bell 2003; Delhaize et al. 2017). The primary input feature is the rest-frame 1.4 GHz radio luminosity, $\log_{10}(L_{1.4\text{GHz}} [\text{W Hz}^{-1}])$. In extended feature sets, we additionally include 3

GHz radio luminosity and redshift to test whether modest increases in radio-related information improve predictive performance. No infrared information is included in any radio-only model.

This conservative feature selection is intentional. Our aim is to quantify the extent to which radio luminosity alone can act as a stand-alone SFR proxy, the implicit assumption behind radio-only SFR catalogues (Condon 1992; Bell 2003; Murphy et al. 2011). Including auxiliary properties such as stellar mass or morphology would change the scientific question from “what does radio encode?” to “what can a multivariate model achieve?”, which is a separate problem. Extended models include at most redshift and a second radio band to test whether modest increases in radio-domain information alter the recoverable variance.

3.2. Machine Learning Models

We implement six supervised regression algorithms using the SCIKIT-LEARN library (Pedregosa et al. 2011), selected to span a wide range of model complexity and inductive bias:

- **Linear Regression**, which tests the assumption of a single global linear relation in log–log space.
- **Support Vector Regression (SVR)** with a radial basis function kernel, which captures smooth non-linear dependencies.
- **k -Nearest Neighbours (kNN)** regression, which probes whether local similarity in radio luminosity corresponds to similar SFRs.
- **Random Forest Regression**, which models hierarchical, non-linear feature interactions and is robust to outliers.
- **Gradient Boosting Regression**, which builds additive corrections to systematic residuals.
- **Multilayer Perceptron (MLP)** neural network, consisting of two hidden layers with 64 and 32 neurons, trained to minimize the mean squared error in log-space.

This ensemble allows us to test whether predictive failure is specific to a particular modelling choice or persists even under highly flexible, non-parametric regression.

3.3. Training and Evaluation

Models are trained using an 80-20 train-test split with a fixed random seed to ensure reproducibility. The split is performed after random shuffling, and we verify that both subsets span the full range of redshift and radio luminosity present in the parent sample. This prevents artificial performance inflation or degradation arising from uneven sampling of parameter space. All models are trained to predict $\log_{10}(\text{SFR})$ and are first evaluated on the held-out COSMOS test set ($N = 1,546$). Model performance is quantified primarily using the coefficient of determination,

$$R^2 = 1 - \frac{\sum (y_{\text{true}} - y_{\text{pred}})^2}{\sum (y_{\text{true}} - \bar{y}_{\text{true}})^2}, \quad (2)$$

computed in log-space (Equation 2). Physically, R^2 measures the fraction of the variance in star formation rate that can be explained by the input radio features alone. All R^2 values reported in this paper are test-set scores unless otherwise stated. For five of the six models (Linear Regression, SVR, kNN, Gradient Boosting, and MLP), training-set R^2 values are within 0.02–0.05 of their test-set counterparts, confirming minimal overfitting. The exception is Random Forest, which achieves $R^2_{\text{train}} = 0.897$ but $R^2_{\text{test}} = 0.198$, indicating severe overfitting when trained on 1.4 GHz luminosity alone; for this reason, Random Forest predictions are treated with caution and the ensemble ceiling is better characterised by the other five models. The consistent performance ceiling across the five well-generalising models confirms that the limit is a genuine informational constraint rather than a model generalisation failure.

To provide a physically meaningful performance baseline, we bootstrap the expected R^2 under a perfect mapping: we draw radio luminosities from the observed full-sample distribution, apply the best-fitting relation, add Gaussian scatter of amplitude σ_{obs} (the residual standard deviation around the best-fit line for the full sample), and fit a linear model to each realisation. The resulting distribution of R^2 values, centred near $R^2_{\text{boot}} \approx 0.85\text{--}0.90$ for the full-sample scatter of $\sigma_{\text{obs}} = 0.51$ dex ($N = 7,729$), represents the performance achievable by a perfect, noise-limited mapping and serves as our fair upper-bound comparison. This value is considerably lower than unity because the intrinsic scatter in the radio–SFR relation is large: even a perfect mapping cannot achieve a high R^2 when the residual noise is this substantial (Yun et al. 2001; Bell 2003). All reported model R^2 values are test-set scores and are placed in context relative to both the full-sample bootstrap baseline ($R^2_{\text{boot}} \approx 0.85\text{--}0.90$) and the test-set perfect-mapping ceiling ($R^2_{\text{perfect}} = 0.70$; see Section 4.2).

To retain physical interpretability, we also compute mean and median absolute errors in linear SFR space, which reflect the typical magnitude of misestimates expected when inferring SFRs from radio data alone.

3.4. AGN and Radio-excess Sources

AGN identified in the COSMOS sample via X-ray, mid-infrared, or SED-based diagnostics are retained in the primary analysis, though results are also examined separately for AGN and non-AGN subsamples. We do not exclude radio-excess sources using infrared–radio ratio cuts. This choice is deliberate. We find that removing radio-excess sources does tighten the observed relation, but this tightening reflects a selection effect rather than a physical improvement in the radio–SFR mapping: the excluded systems are a morphologically and physically heterogeneous population that cannot be uniformly classified as AGN from radio data alone.

As detailed in Section 2.4, to convert luminosities to flux densities we adopted a flat Λ CDM cosmology with parameters from the Planck Collaboration (Planck Collaboration et al. 2020) results: $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_m = 0.315$. For each cluster at redshift z , we compute the luminosity distance $D_L(z)$ using Astropy’s cosmology utilities (Astropy Collaboration et al. 2013, 2018).

As detailed in Section 2.5, radio-excess hosts are compact, smooth, and concentrated, properties associated with early-type galaxies (van der Wel et al. 2014), rather than with the disturbed or star-forming morphologies expected if the excess arose from merger-driven AGN triggering (Holt et al. 2008; O’Dea & Saikia 2021). Their exclusion therefore removes a physically real component of the galaxy population without robust justification, and would not be possible in a uniform way across the heterogeneous multi-wavelength coverage of wide-field radio surveys.

By retaining radio-excess sources, we evaluate the reliability of radio continuum emission as a star formation tracer under realistic observational conditions directly relevant to MeerKAT and forthcoming SKA surveys.

3.5. Application to Cluster Environments

Models trained on the COSMOS field sample are applied without retraining to the MGCLS cluster data. This transfer tests whether any radio–SFR relation learned in the field generalizes to dense environments, where additional non-star-forming radio emission from AGN activity and relativistic electrons in the intra-cluster medium is expected (Croston et al. 2019; Cuciti et al. 2021).

For cluster sources, rest-frame radio luminosities are computed assuming the cluster redshift. Sources are further divided by projected cluster-centric distance relative to R_{500} to assess environmental dependence. This approach allows us to test not only whether radio luminosity predicts star formation, but whether any predictive structure is robust to environmental complexity. Where flux densities are required for direct comparison with observations, luminosities are converted using $S_{\nu_{\text{obs}}} = (L_\nu / 4\pi D_L^2)(1+z)^{1-\alpha}$ with $\alpha = -0.7$ (Condon 1992); flux densities are quoted in mJy throughout. Although assuming the cluster redshift for all sources introduces uncertainties for foreground and background galaxies, the same qualitative behaviour is observed in the COSMOS field where source-specific redshifts are available.

3.6.

4. RESULTS

We now quantify the extent to which 1.4 GHz radio luminosity encodes information about star formation rate (SFR). Infrared-derived SFRs are treated as the reference measure of star formation activity. Our objective is not to refine a calibration, but to determine the extent to which radio luminosity can reliably recover SFR under realistic observational conditions.

4.1. Global Radio Luminosity–Star Formation Rate Relation

We begin by examining the empirical relation between rest-frame 1.4 GHz radio luminosity and infrared-derived SFR in the VLA–COSMOS sample, shown in Figure 1. The relation is approximately linear in log–log space, but exhibits a slope shallower than unity ($m < 1$) and substantial intrinsic scatter (Yun et al. 2001; Bell 2003; Delhaize et al. 2017). We fit a linear relation in log–log space to the non-AGN population and find a best-fit slope of $m = 0.577 \pm 0.018$ on the test set. The Pearson correlation coefficient is $r = 0.659$, implying that $\sim 43\%$ of the variance in SFR is linearly associated with radio luminosity (i.e.,

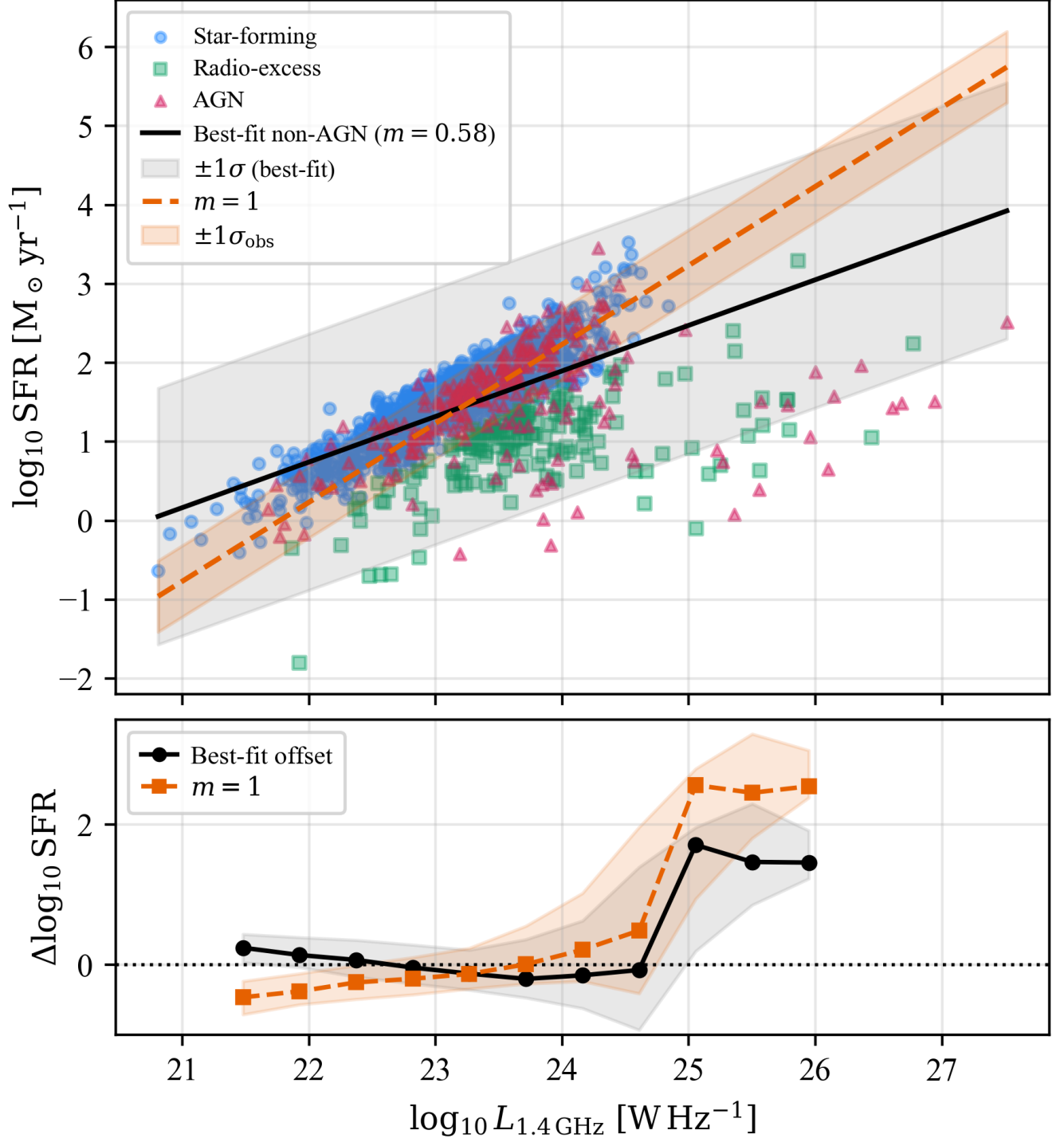


Figure 1. *Top:* Rest-frame 1.4 GHz radio luminosity versus infrared-derived SFR for the matched COSMOS test set ($N = 1,546$; $z = 0.10\text{--}5.37$), separated into star-forming galaxies (blue circles), radio-excess sources (green squares), and AGN (red triangles). The solid black line shows the best-fit relation excluding AGN ($m = 0.577 \pm 0.018$; Pearson $r = 0.659$), with the grey shaded region indicating the $\pm 1\sigma$ scatter ($\sigma_{\text{obs}} = 0.45$ dex). The orange dashed line shows the *unity slope*: slope $m = 1$, anchored at the sample pivot (mean $\log L_{1.4}$, mean $\log \text{SFR}$), whose $\pm \sigma_{\text{obs}}$ scatter band (orange shading) represents the noise-limited ceiling for any unity-slope calibration. Its theoretical R^2 on this test set is $R^2_{\text{perfect}} = \sigma_x^2 / (\sigma_x^2 + \sigma_{\text{obs}}^2) = 0.70$ (with $\sigma_x = 0.68$ dex, $\sigma_{\text{obs}} = 0.45$ dex). *Bottom:* Systematic offset $\Delta \log_{10}(\text{SFR}) = \text{predicted} - \text{true}$, binned with median $\pm 1\sigma$ shading, for both calibrations. The black line (best-fit, $m = 0.577$) curves upward at $\log L > 25$ due to radio-excess and AGN sources. The orange dashed line (perfect proportional mapping, $m = 1$) shows the opposite systematic at low luminosities, underestimating SFR by ~ 0.4 dex below $\log L \sim 22.5$. Neither calibration avoids luminosity-dependent biases at extreme luminosities, demonstrating that the sub-unity slope is intrinsic to the radio–SFR relation rather than an artefact of sample selection. The slope of $m = 0.577 \pm 0.018$ (Pearson $r = 0.659$), significantly shallower than unity, implies systematic overestimates at high $L_{1.4}$ and underestimates at low $L_{1.4}$ for any unity-slope calibration, with offsets exceeding 0.3 dex at extreme luminosities.

$r^2 \approx 0.43$). We emphasize that Pearson r is a descriptive measure of monotonic association, whereas the R^2 values reported below are obtained from explicit predictive models evaluated on held-out data (Section 3.3).

AGN hosts exhibit a flatter relation ($m \simeq 0.27$), but they do not dominate the overall dispersion. Importantly, the majority of AGN host galaxies and radio-excess sources lie within 3σ of the best-fitting radio luminosity–SFR relation defined by the non-AGN population. These systems therefore do not constitute extreme outliers that artificially inflate the observed scatter, but instead populate the same broad distribution as star-forming galaxies. The weakness and shallow slope of the radio–SFR relation cannot therefore be attributed to a small subset of pathological AGN-dominated sources; rather, the scatter reflects intrinsic physical variability in the connection between radio emission and star formation (Lacki et al. 2010; Murphy 2009; Schleicher & Beck 2016).

A slope shallower than unity implies a luminosity-dependent bias when a unity-slope calibration is assumed. For $m = 0.577 \pm 0.018$, this corresponds to systematic errors of ~ 0.5 dex per decade in $L_{1.4}$ relative to the pivot luminosity: a source one decade above the pivot has its SFR overestimated by a factor of ~ 3.2 , while a

source one decade below is underestimated by a similar factor. Two decades away the bias approaches ~ 0.8 – 1.0 dex. The lower panel of Figure 1 illustrates this directly, and the mathematical origin of these biases is discussed further in Section 5.2. These biases persist even after excluding AGN-classified sources, confirming they are intrinsic to the radio–SFR mapping rather than an artefact of contamination. We note that adopting the best-fit slope ($m = 0.577$) rather than unity removes the leading luminosity-dependent term and so substantially reduces the calibration bias, but it does not eliminate it: the residual scatter remains asymmetric and luminosity-dependent because the off-relation radio-excess and AGN population steepens the apparent relation at the bright end (lower panel of Figure 1).

4.2. Perfect Mapping Baseline and Variance Decomposition

To place the predictive performance of our models in a physically meaningful context, we construct a *perfect proportional mapping*: slope $m = 1$, anchored at the sample pivot (mean $\log_{10} L_{1.4\text{ GHz}} = 23.27$, mean $\log_{10} \text{SFR} = 1.46$ on the test set), and subject only to the observed scatter $\sigma_{\text{obs}} = 0.45$

dex. This represents the calibration implicitly assumed by Condon (1992) and Bell (2003): a universal proportional (unity-slope) mapping between radio luminosity and SFR. Its theoretical R^2 is

$$R_{\text{perfect}}^2 = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_{\text{obs}}^2} = \frac{0.68^2}{0.68^2 + 0.45^2} = 0.70, \quad (3)$$

where $\sigma_x = 0.68$ dex is the standard deviation in $\log_{10} L_{1.4\text{ GHz}}$ among non-AGN sources in the test set. This baseline rests on two explicit assumptions, which we state so that the decomposition can be assessed directly. First, we take the residual scatter σ_{obs} measured about the best-fit non-AGN relation as the relevant irreducible noise term; this is appropriate if that scatter is dominated by genuine source-to-source variation in synchrotron emissivity rather than by measurement error or by the slope itself, and we treat it as an upper bound precisely because it cannot fully exclude those other contributions (Section 5.1).

Second, we adopt the pivot-anchored unity-slope line as the idealisation of the canonical proportional calibration, which fixes the normalisation at the sample mean rather than at any external zero-point; this isolates the slope as the only degree of freedom distinguishing the canonical calibration from a free-slope fit, and is the construction under which R_{perfect}^2 depends only on the ratio $\sigma_x/\sigma_{\text{obs}}$ (Equation 3) and not on the absolute calibration zero-point. Under these two assumptions the decomposition below follows directly; relaxing either would change the split between the noise-floor and calibration-bias terms but not the central finding that the model ceiling lies well below the noise-limited limit.

This result has two immediate implications. First, even if the calibration were perfect in every other respect (zero model error, no sample selection, no AGN), it would still explain only 70% of the variance in SFR on this test set, because σ_{obs} is physically large. Second, the gap between the model ceiling ($R^2 \lesssim 0.45$) and this perfect-mapping limit ($R^2 = 0.70$) can itself be decomposed into two physically distinct contributions, summarized in Table 5.

The noise-floor contribution ($\Delta R^2 = 0.30$) is irreducible: it reflects genuine physical variability in synchrotron emissivity across the galaxy population that cannot be inferred from radio luminosity alone. The calibration-bias contribution ($\Delta R^2 = 0.26$) is in principle avoidable but only if the sub-unity slope is explicitly modelled. Crucially, these two contributions are of comparable magnitude, placing slope-refinement on equal footing with understanding the physical scatter as a priority for calibration improvement. This quantification has not previously been reported in the literature.

Table 5. Decomposition of the unrecovered SFR variance on the COSMOS test set ($N = 1,546$). The perfect proportional mapping ($m = 1$, pivot-anchored, scatter $\sigma_{\text{obs}} = 0.45$ dex) sets the noise-limited ceiling $R_{\text{perfect}}^2 = 0.70$ (Equation 3); the best radio-only models saturate at $R_{\text{models}}^2 \lesssim 0.45$. The total unrecovered variance ($1 - R_{\text{models}}^2 \approx 0.55\text{--}0.57$) splits into an irreducible physical noise floor and an avoidable calibration bias arising from the sub-unity slope.

Component	ΔR^2	Reducible?
Perfect-mapping ceiling, R_{perfect}^2	0.70	...
Model ceiling, R_{models}^2	$\lesssim 0.45$	...
Calibration bias (sub-unity slope, $m = 0.577$)	0.26	Yes, if slope modelled
Irreducible noise floor (physical scatter)	0.30	No

NOTE—The two reducible/irreducible contributions are of comparable magnitude. The calibration-bias term is recoverable in principle by fitting a free slope rather than assuming $m = 1$; the noise-floor term reflects intrinsic variation in magnetic field strength, cosmic-ray transport, and star formation history that cannot be inferred from radio luminosity alone.

To establish where the calibration-bias contribution physically originates, we recompute the empirical fit and the perfect-mapping baseline for a series of subsamples defined by an upper cut in radio luminosity, retaining only non-AGN sources with $\log_{10} L_{1.4\text{ GHz}} < L_{\text{cut}}$ (Figure 2 and Table 6). As the bright end is progressively removed, the best-fit slope steepens from $m = 0.577$ toward $m = 0.739$ and the empirical R^2 rises from 0.43 to 0.56, while the perfect-mapping ceiling R_{perfect}^2 remains pinned near 0.70 across all cuts. The slope-bias contribution $\Delta R^2 = R_{\text{perfect}}^2 - R_{\text{empirical}}^2$ correspondingly falls from 0.26 to 0.14. The lower panels of Figure 2 make the behaviour explicit: at the two faintest cuts the systematic offset of the best-fit relation flattens to within ~ 0.1 dex across the entire retained luminosity range, whereas at the most permissive cut ($L_{\text{cut}} < 26$) both the best-fit and proportional-mapping offsets diverge sharply upward above $\log_{10} L_{1.4\text{ GHz}} \approx 24.5$. This isolates the sub-unity slope as a property of the high-luminosity population, namely the radio-excess and AGN sources whose true SFRs turn over at $\log_{10} L_{1.4\text{ GHz}} \gtrsim 25$ (lower panel of Figure 1), rather than a uniform feature of the relation.

Taken at face value, this experiment appears to offer a simple remedy: a bright-end luminosity cut restores a tight, low-bias radio–SFR relation without any multi-wavelength information. We caution against this interpretation for three reasons. First, the cut is not independent of the q_{IR} radio-excess criterion: as L_{cut} tightens from none to < 24 , the retained non-AGN count falls from 170 to 121 (Table 6), so the truncation removes the same morphologically distinct, off-relation population that the q_{IR} cut targets (Section 2.5). The two procedures are therefore not independent validations of one another; both operate by discarding the same bright, radio-excess sources. Second, the cut cannot be applied prospectively. Identifying which high-luminosity sources lie off the relation requires knowing their SFRs in advance, which is precisely the quantity being inferred; in a blind survey there is no radio-only criterion that isolates the off-relation population, because (as Figure 2 shows) the radio-excess sources do not form a detached cloud but track the relation with a modest, coherent downward offset and overlap the star-forming locus throughout. Third, and most fundamentally, even the most aggressive cut leaves the empirical R^2 saturating at 0.56, well short of the 0.70 ceiling: the residual slope bias plus the irreducible noise floor persists after the bright tail is removed, confirming that the variance limitation is not an artefact of the luminosity range probed.

We note that the coherent, modest nature of the radio-excess offset admits an alternative reading: rather than discarding these sources, one could in principle adopt a two-component calibration, with a separate offset relation for the radio-excess population. Such a scheme would retain the sources while accounting for their elevated radio emission per unit SFR. It does not, however, evade the central difficulty, since it still requires the radio-excess population to be identified before the calibration can be applied, and that identification cannot be made from radio luminosity alone (Section 2.5). A population-aware calibration is thus a promising direction, but one that depends on exactly the resolved multi-wavelength or morphological classification that wide-field radio surveys will generally lack.

Table 6. Evolution of the radio-SFR relation under progressive upper cuts in radio luminosity. Each row retains only non-AGN test-set sources with $\log_{10}(L_{1.4\text{ GHz}}/\text{W Hz}^{-1}) < L_{\text{cut}}$, removing the bright end. The empirical slope m and empirical R^2 are from a free-slope linear fit; $R_{\text{perfect}}^2 = \sigma_x^2/(\sigma_x^2 + \sigma_{\text{obs}}^2)$ is the noise-limited ceiling; $\Delta R^2 = R_{\text{perfect}}^2 - R_{\text{empirical}}^2$ is the slope-bias gap.

L_{cut}	N	m	$R_{\text{empirical}}^2$	R_{perfect}^2	ΔR^2
none	1546	0.577 ± 0.018	0.434	0.697	0.263
< 26	1537	0.598 ± 0.018	0.448	0.695	0.247
< 25	1514	0.695 ± 0.018	0.535	0.704	0.169
< 24	1364	0.739 ± 0.019	0.556	0.697	0.140

NOTE—As the bright end is excised, the empirical slope climbs from 0.58 toward ~ 0.74 and the empirical R^2 rises from 0.43 to 0.56, while R_{perfect}^2 remains stable at ~ 0.70 . The slope-bias gap ΔR^2 nearly halves, demonstrating that the sub-unity slope is driven by the high-luminosity population. The decreasing non-AGN count reflects the loss of radio-excess sources, which preferentially occupy the bright end.

We further note that the perfect-mapping ceiling is itself stable under this truncation: although σ_{obs} falls from 0.45 to 0.37 dex as the bright tail is excised, σ_x decreases in lockstep (from 0.68 to 0.56 dex), so the ratio in Equation 3 is conserved and R_{perfect}^2 does not depend on where the luminosity window is placed. We note that the full-sample bootstrap baseline ($R_{\text{boot}}^2 \approx 0.85\text{--}0.90$; $\sigma_{\text{obs}} = 0.51$ dex; $N = 7,729$) differs from the test-set perfect-mapping ceiling ($R_{\text{perfect}}^2 = 0.70$; $\sigma_{\text{obs}} = 0.45$ dex; $N = 1,546$) because the two quantities refer to different samples. The bootstrap was computed using the wider luminosity dynamic range of the full COSMOS dataset ($\sigma_x \approx 1.0$ dex), while the test-set value reflects the narrower luminosity spread of the held-out subset ($\sigma_x = 0.68$ dex). Both are correct; they quantify the noise-limited ceiling under their respective sample conditions. All reported model R^2 values are test-set scores and should be compared to $R_{\text{perfect}}^2 = 0.70$ as the operationally relevant baseline.

4.3. Machine Learning Recovery of Star Formation Rates

We next assess predictive performance using supervised machine learning models trained on radio-only inputs. The test-set coefficients of determination for all six models and feature sets are summarised in Table 7. Across all models, mean absolute errors exceed several hundred $\text{M}_{\odot} \text{ yr}^{-1}$, corresponding to typical uncertainties of $\gtrsim 0.5\text{--}1.0$ dex. The best-performing algorithms achieve $R^2 \lesssim 0.45$ in log-space, meaning that more than half of the intrinsic variance in SFR is not captured by radio luminosity alone. Across all six models, predicted values scatter broadly around the 1:1 relation, with no model recovering the high-SFR tail reliably. A notable feature shared by several models is an apparent upper cap on predicted SFR values. This cap is not the same as the variance ceiling discussed in Section 5: it arises from regression-to-the-mean. When the input feature (radio luminosity) carries limited predictive information, models converge towards predictions near the training-set mean, producing an effective ceiling on predicted values at approximately the training mean plus a small multiple of the training standard deviation. This is a direct consequence of limited information content in the predictor, and does not indicate a physical limit in the underlying SFR distribution.

To assess whether this performance is consistent with a physically reasonable noise floor, we compare the test-set model performance to the perfect-mapping ceiling established in Section 4.2. The gap between $R_{\text{perfect}}^2 = 0.70$ and $R_{\text{models}}^2 \lesssim 0.45$ is $\Delta R^2 \approx 0.26$, attributable to the sub-unity slope bias (Table 5). This means that even a model that learned the empirical slope perfectly and applied it without error would still fail to reach the noise-limited ceiling, because unity-slope calibrations (which the models effectively approximate under regression-to-the-mean) assume $m = 1$ rather than $m = 0.577$. Separating AGN and non-AGN populations (Figure 3) does not

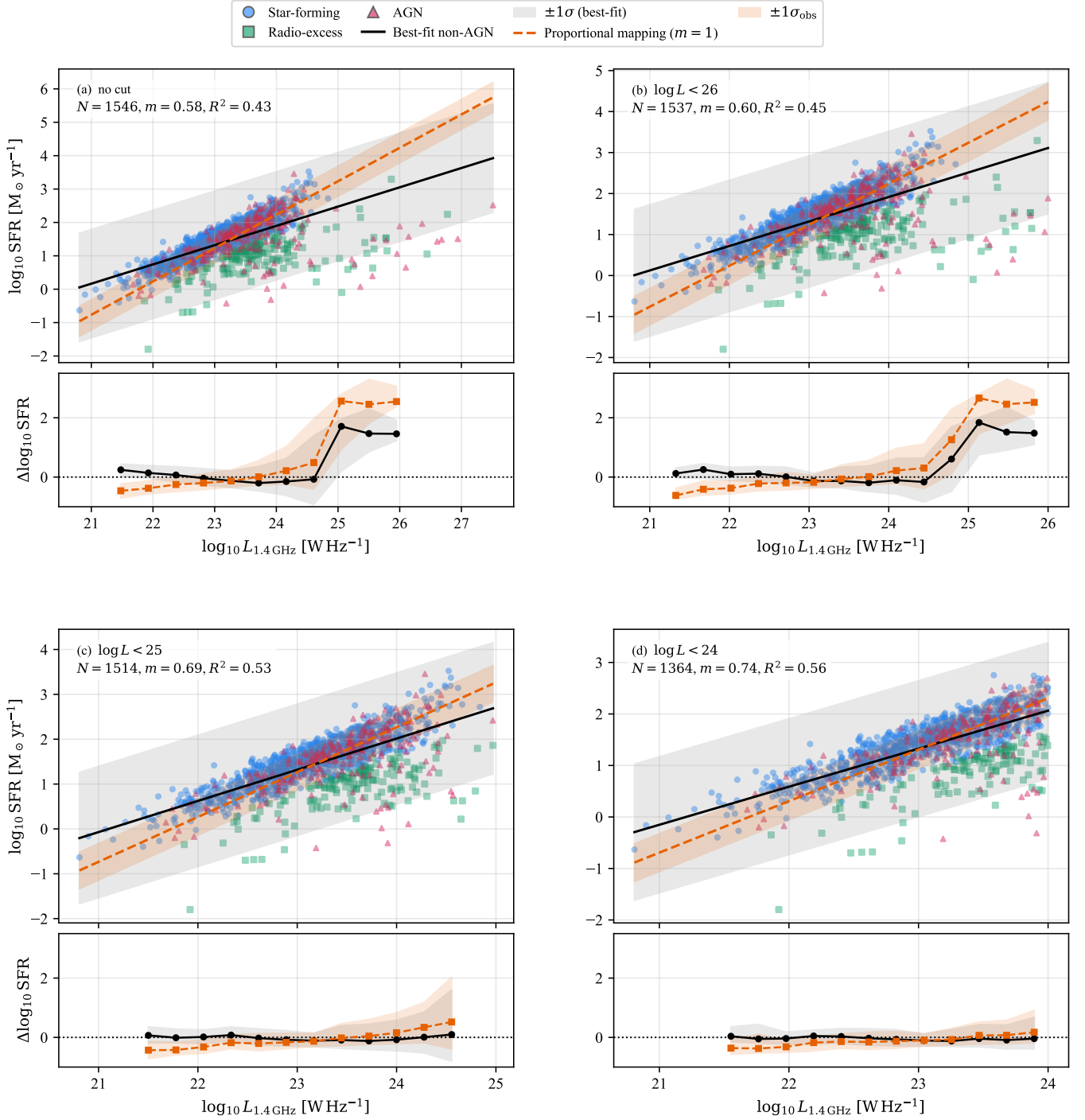


Figure 2. Evolution of the radio–SFR relation under progressive upper cuts in 1.4 GHz radio luminosity, for the COSMOS test set. Panels show (a) the full sample (no cut), (b) $\log_{10} L_{1.4\text{ GHz}} < 26$, (c) < 25 , and (d) < 24 . In each panel the upper plot shows infrared-derived SFR against radio luminosity for star-forming galaxies (blue circles), radio-excess sources (green squares), and AGN (red triangles); the solid black line is the free-slope best fit to the non-AGN population (with $\pm 1\sigma$ scatter, grey band) and the orange dashed line is the unity mapping ($m = 1$, pivot-anchored, with $\pm 1\sigma_{\text{obs}}$ band). The lower plot in each panel shows the binned systematic offset $\Delta \log_{10}(\text{SFR}) = \text{predicted} - \text{true}$ (median with 16th–84th percentile shading) for both calibrations. As the bright end is removed the best-fit slope steepens ($m = 0.58 \rightarrow 0.74$) and the empirical R^2 rises ($0.43 \rightarrow 0.56$); at the two faintest cuts the best-fit offset flattens to within ~ 0.1 dex across the retained range, whereas at $L_{\text{cut}} < 26$ both offsets diverge sharply above $\log_{10} L_{1.4\text{ GHz}} \approx 24.5$. The radio-excess sources (green) do not form a detached outlier cloud: they track the relation with a modest, coherent downward offset and overlap the star-forming locus throughout, which is precisely why they cannot be isolated from radio luminosity alone. The apparent tightening upon truncation is therefore driven by the removal of the same off-relation population targeted by the q_{IR} cut (Section 2.5), and does not constitute an independent, prospectively applicable remedy (Table 6).

significantly improve predictive performance, confirming that the shortfall is not attributable to AGN contamination of the training set.

Table 7. Average coefficient of determination (R^2) for regression models trained using different input feature sets. Values are computed in $\log_{10}(\text{SFR})$ space and evaluated on the held-out test set ($N = 1,546$). The test-set perfect-mapping ceiling is $R^2_{\text{perfect}} = 0.70$.

Model	Model R^2		
	1.4 GHz Only	Radio-only	Radio + WISE
Linear Regression	0.33	0.34	1.00
Support Vector Regression	0.46	0.47	0.98
k -Nearest Neighbors	0.36	0.39	0.99
Random Forest	0.20	0.27	1.00
Gradient Boosting	0.45	0.46	1.00
Multilayer Perceptron	0.44	0.47	0.99
Mean R^2	0.37	0.40	0.99

Notes: Radio-only models consistently saturate at $R^2 \lesssim 0.45$, compared to the test-set perfect-mapping ceiling of $R^2_{\text{perfect}} = 0.70$. The gap ($\Delta R^2 \approx 0.26$) reflects the sub-unity slope bias (Table 5); the remaining deficit below R^2_{perfect} ($\Delta R^2 \approx 0.30$) is the irreducible noise floor. The near-unity R^2 values in the Radio+WISE case reflect the inclusion of infrared information directly correlated with the target variable and therefore represent a consistency check rather than an independent predictive experiment.

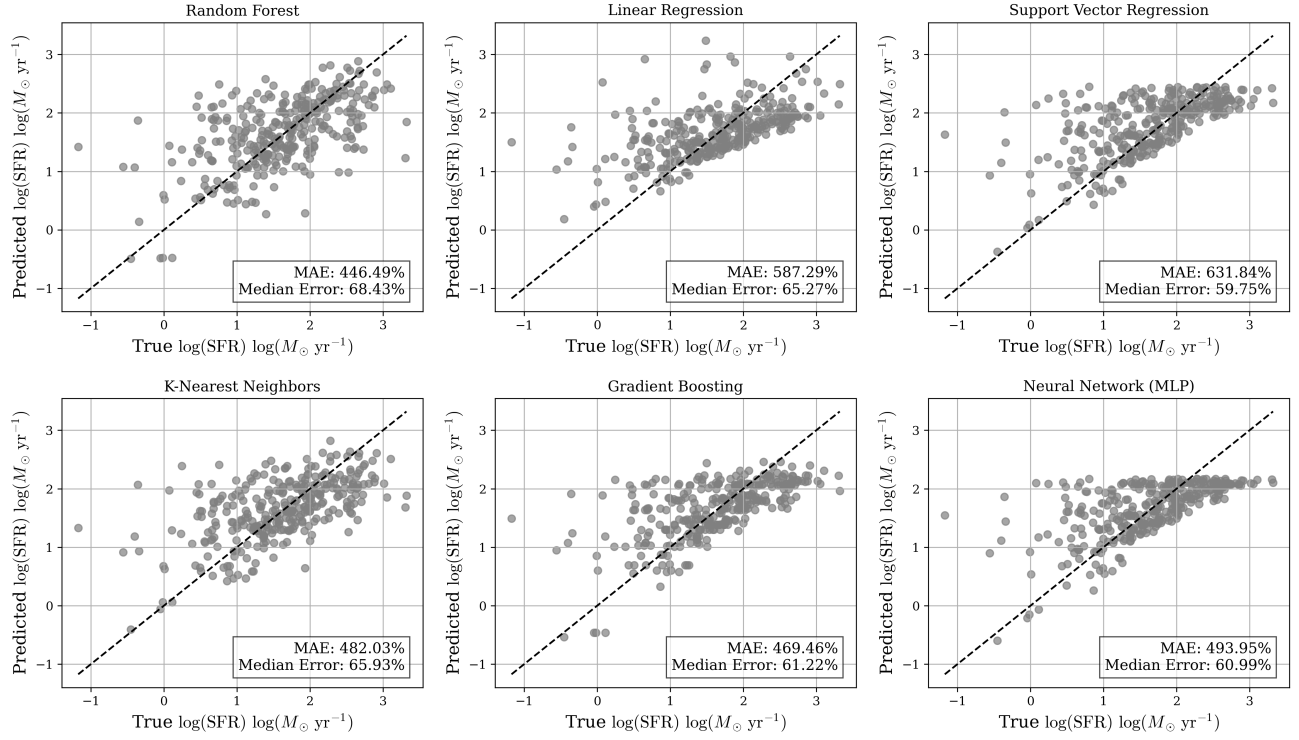
4.4. Environmental Dependence in MGCLS Clusters

Figure 4 shows predicted versus observed 1.4 GHz flux densities (in mJy) for both the cluster and field samples. To identify the flux density at which the predicted-to-observed flux ratio changes behaviour, marking the empirical transition from star-formation-dominated to AGN- or environment-dominated radio emission, we apply Gaussian Mixture Modelling (GMM) to the joint distribution of observed flux density and flux ratio for each cluster. GMM fits a two-component mixture to this distribution in log space, and the midpoint between the two component means defines the breakpoint. Spearman rank correlations between observed flux density and the predicted-to-observed ratio are computed in bins to quantify how correlation strength varies across the flux range. Figure 5 summarizes the resulting breakpoints and Spearman correlations across the full cluster sample. The cluster sample delivers a result that the COSMOS field cannot: a direct measurement of how the predictive reliability of radio luminosity varies with environment. Across all clusters, no strong correlation is observed between predicted and measured flux densities, consistent with the COSMOS ceiling and confirming that the field result transfers to dense environments. The environment-specific finding, however, is the systematic radial trend. Outside the cluster cores, the predicted-to-observed correlation is $\rho \sim 0.39$, comparable to the field; within R_{500} it falls to $\rho \sim 0.21$, a degradation of nearly a factor of two (Figure 5).

This decline is not present in the COSMOS field and cannot be recovered from it: it isolates the additional, environmentally driven loss of radio-SFR predictability in cluster cores, where intracluster synchrotron emission, shocks, and ram-pressure-affected sources contribute radio flux unrelated to ongoing star formation (Croston et al. 2019; Cuciti et al. 2021). The cluster sample therefore contributes a distinct physical result, namely that dense environments degrade radio-based SFR inference beyond the field-level ceiling, rather than merely reproducing the COSMOS answer.

We emphasize that the ~ 0.2 mJy breakpoint identified via Gaussian Mixture Modelling represents an empirical transition in predictive behaviour rather than a sharp physical threshold. It marks the flux regime where additional sources of variance, such as AGN activity and cluster-related synchrotron processes, begin to dominate over star-formation-driven emission (Padovani et al. 2015; Croston et al. 2019). An illustrative example is shown in Figure 7 (Appendix A) for Abell 22, and Figure 6 shows the AGN contribution to GMM-identified populations and the AGN fraction per cluster.

Predicted vs True SFR (AGN)



Predicted vs True SFR (non-AGN)

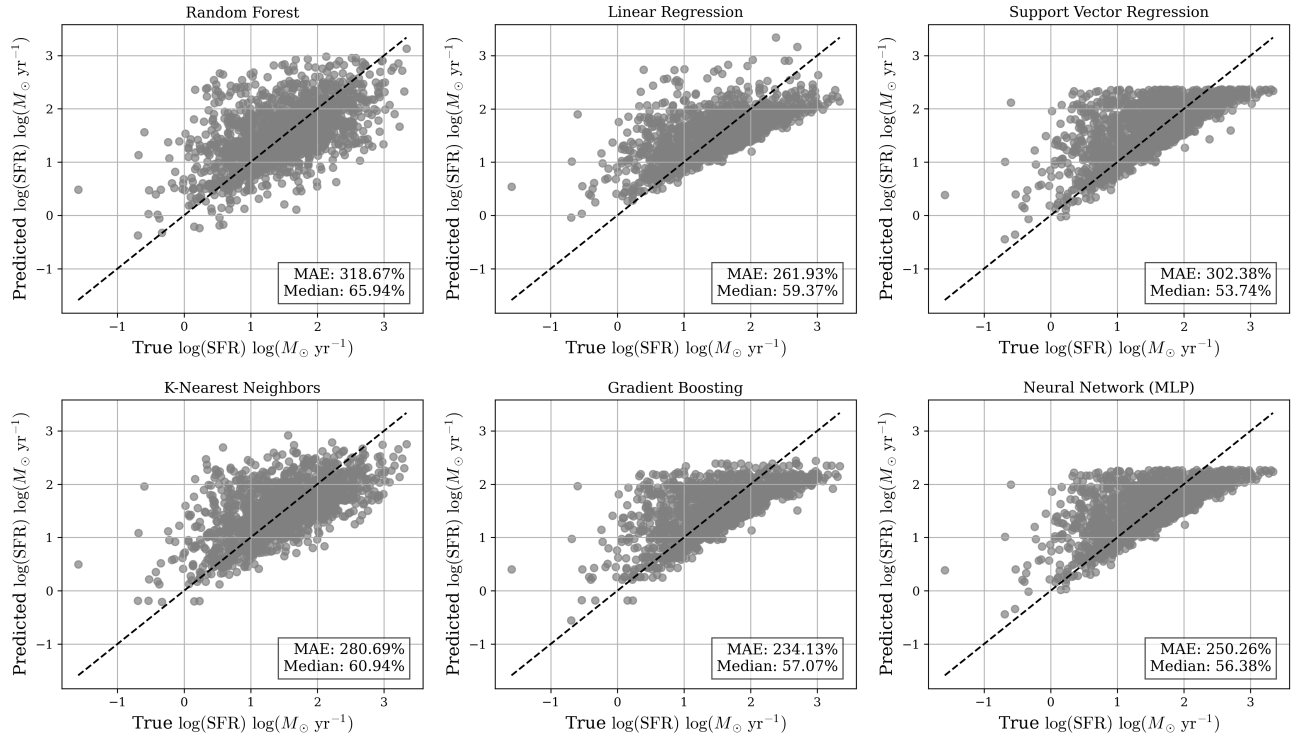


Figure 3. Predicted versus true SFRs for AGN hosts (top) and non-AGN galaxies (bottom). Significant scatter persists even after AGN removal, demonstrating that the breakdown of the radio–SFR relation is not driven solely by AGN contamination (Lacki et al. 2010; Murphy 2009).

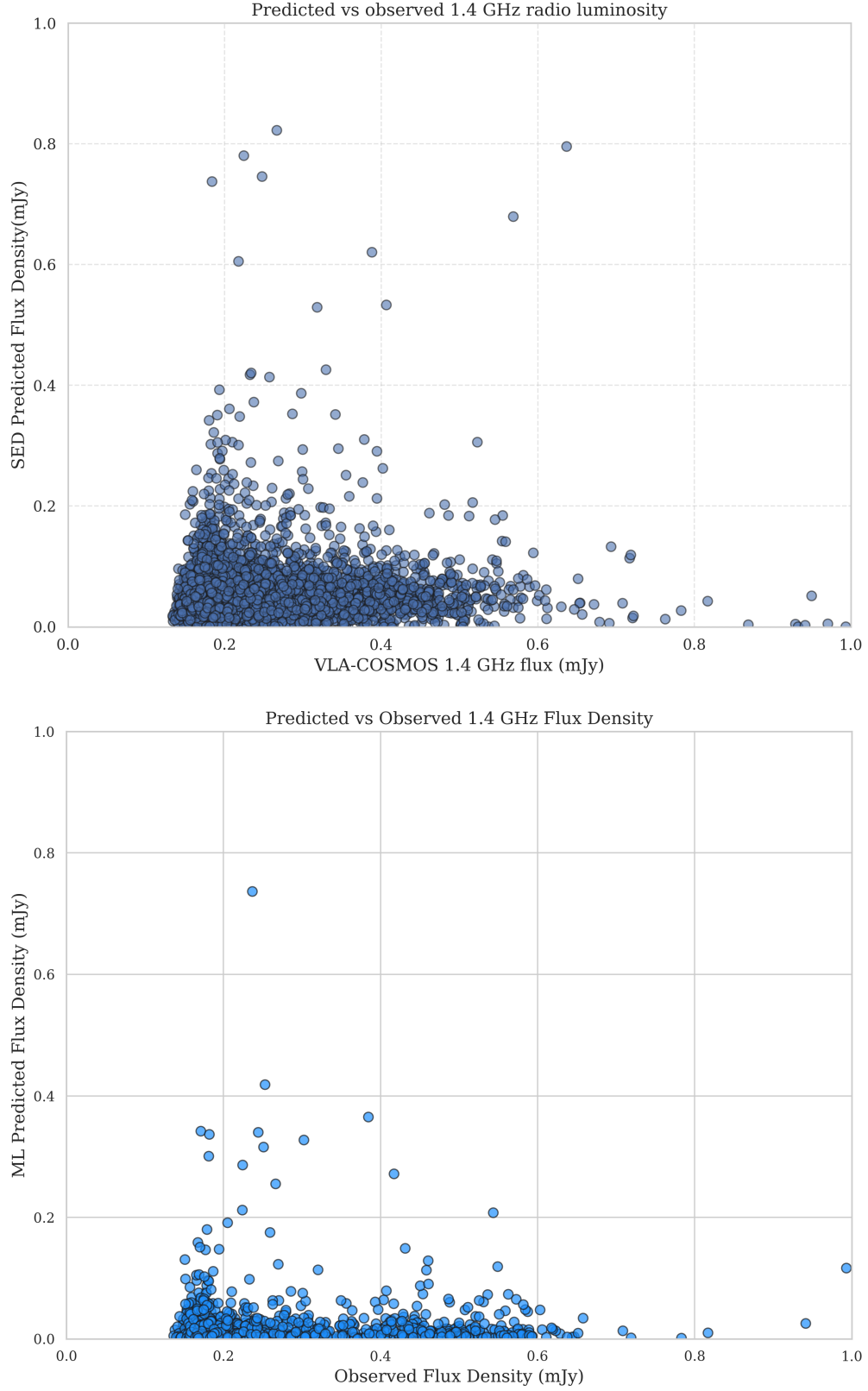


Figure 4. Comparison of predicted and observed 1.4 GHz flux densities (in mJy) in clusters (top) and the COSMOS field (bottom). No strong correlation is observed in either environment, consistent with the broad scatter in the radio–SFR relation (Yun et al. 2001; Delhaize et al. 2017).

Breakpoints Comparison (Mean vs Median)

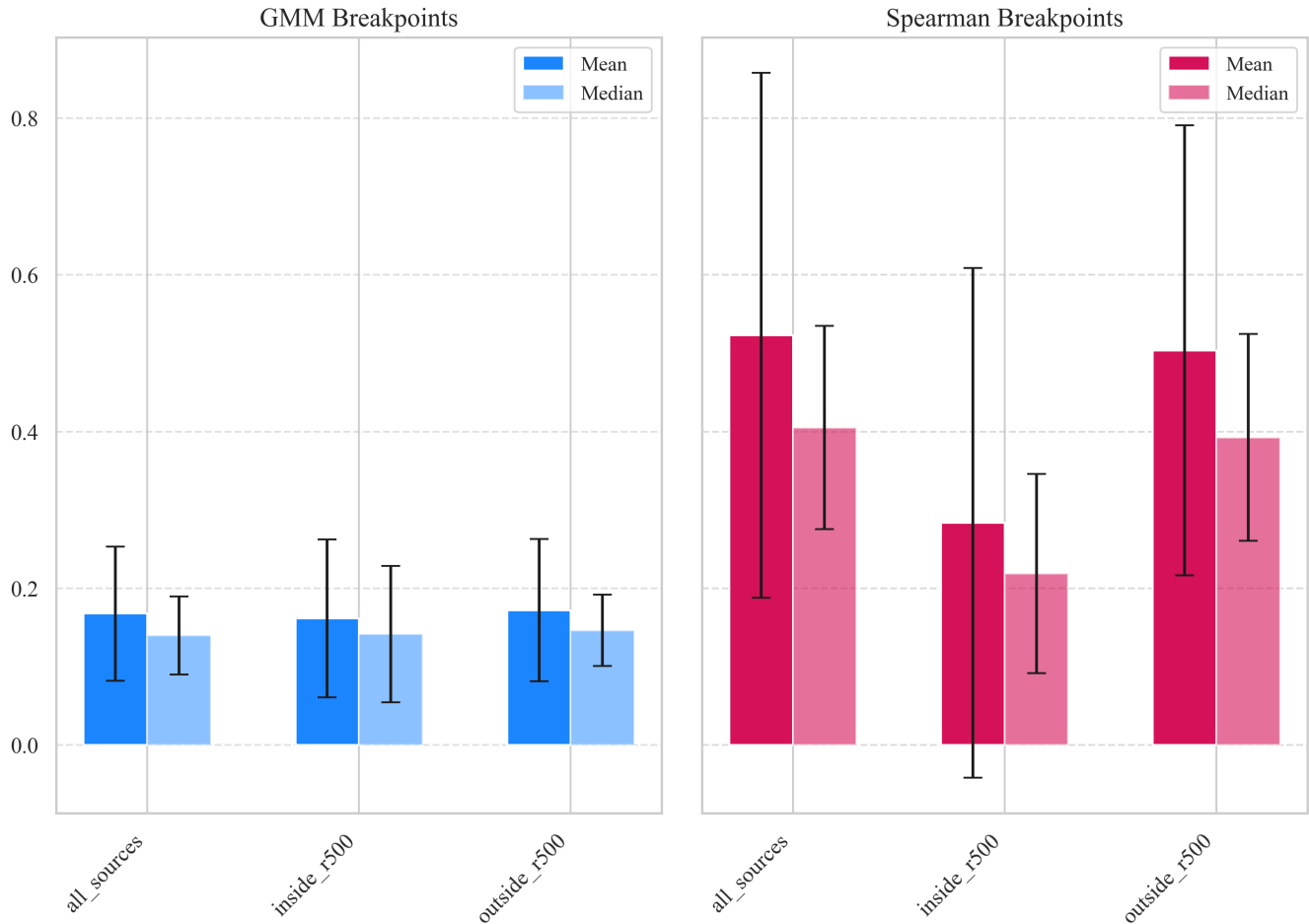


Figure 5. Mean and median Gaussian Mixture Model breakpoints and Spearman correlations for all sources, inside R_{500} , and outside R_{500} . Correlations are systematically weaker within cluster cores, consistent with environmental modulation of radio emission (Croston et al. 2019; Cuciti et al. 2021).

4.5. Radio-excess Sources and AGN Contribution

As shown in Figure 6, moderate radio excess ($\lesssim 0.5$ dex below the mean IR–radio relation) is observed in systems without independent AGN classifications. Removing such systems does restore a substantially tighter radio–SFR relation. However, as argued in Section 2.5, the physical nature of these sources is ambiguous without resolved morphological or spectroscopic data, and their compact, smooth morphologies (Table 3) are inconsistent with a simple AGN interpretation. We therefore retain them and note that their inclusion produces a more conservative and realistic characterisation of radio-based SFR inference. The persistence of weak correlations after masking only the independently identified AGN, without removing the broader radio-excess population, confirms that the radio-excess sources themselves carry significant predictive uncertainty, and that their classification is not straightforward from radio data alone.

4.6.

5. DISCUSSION

While previous studies have characterized the intrinsic scatter of the infrared–radio relation (e.g., Yun et al. 2001; Bell 2003; Delhaize et al. 2017), our analysis quantifies the variance in SFR that is recoverable from radio luminos-

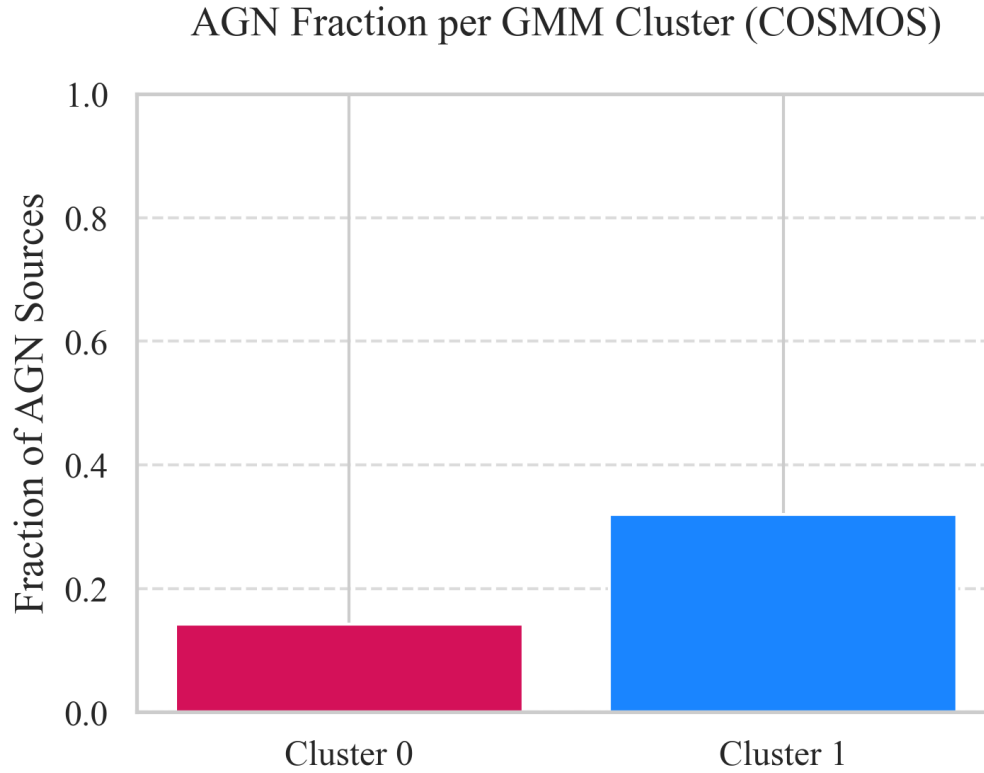
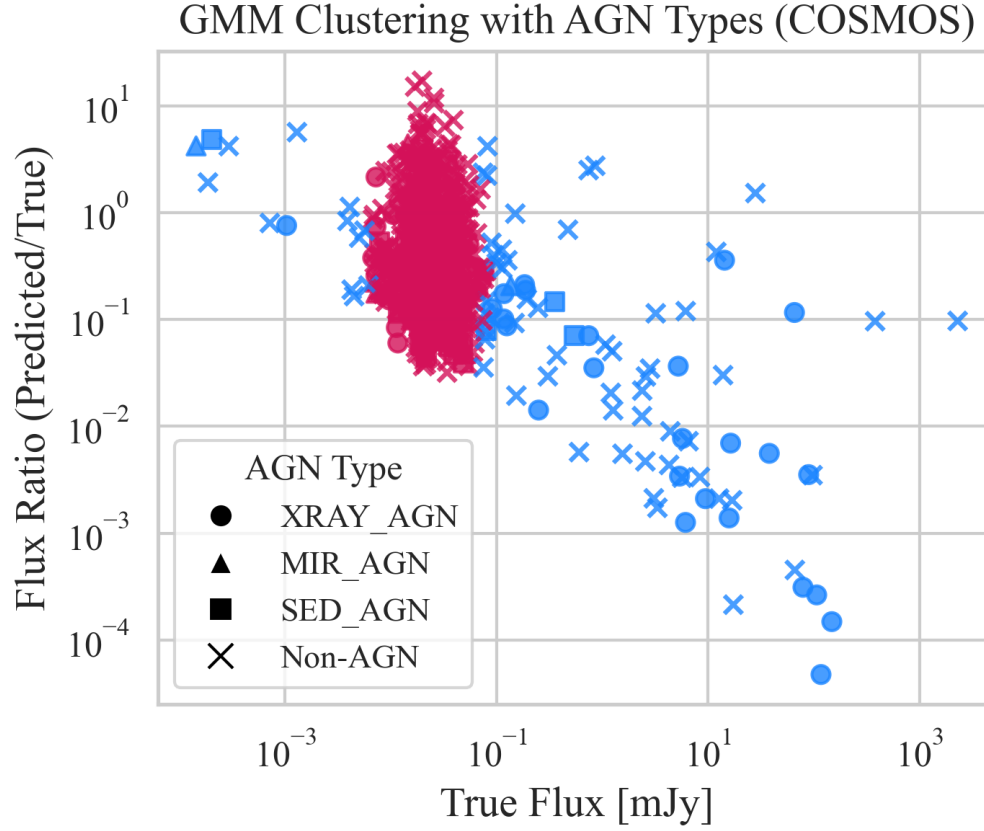


Figure 6. AGN contribution to GMM-identified populations (top) and AGN fraction per cluster (bottom). While AGN contribute to the high-luminosity tail, a substantial fraction of radio-excess sources do not carry independent AGN classifications and show morphological properties more consistent with early-type galaxies than with AGN-driven systems (Best et al. 2005; Hardcastle & Croston 2020).

ity alone. The convergence of predictive performance across linear and non-linear models indicates that the $\lesssim 45\%$ variance ceiling reflects the information content of the radio features we test, rather than limitations of a specific regression architecture. This convergence across Linear Regression, Random Forest, Support Vector Regression, Gradient Boosting, and Multilayer Perceptron models demonstrates that increasing model flexibility does not recover additional predictive power; the limitation is therefore set by the features rather than the fitting method. We emphasise that this is an empirical, feature-specific ceiling demonstrated by cross-model agreement, not a formal information-theoretic bound: observables beyond radio luminosity, redshift, and a second radio band, in particular resolved morphology (Section 5.1), may carry additional SFR information that these inputs do not.

5.1. The Physical Origin of the Variance Ceiling

The maximum predictive performance achieved by radio-only models ($R^2 \lesssim 0.45$) implies that more than half of the intrinsic variance in SFR is not encoded in radio luminosity alone. To place this in context, we constructed a perfect proportional mapping anchored at the sample pivot with slope $m = 1$ (the unity-slope calibration of Condon 1992; Bell 2003), which achieves a theoretical R^2 of 0.70 on the test set (Section 4.2). The total unrecovered variance ($1 - R_{\text{models}}^2 \approx 0.57$) decomposes cleanly as (Table 5):

- An *irreducible noise floor* ($\Delta R^2 = 0.30$), arising from genuine physical variability in synchrotron emissivity (magnetic field strength, cosmic-ray confinement efficiency, and time-varying star formation histories) that no model can recover from radio luminosity alone.
- A *calibration bias* ($\Delta R^2 = 0.26$), arising from the sub-unity slope ($m = 0.577$): any unity-slope calibration that assumes $m = 1$ introduces this additional variance loss on top of the noise floor. It is in principle avoidable, but only if the slope is explicitly modelled rather than assumed.

These two contributions are of comparable magnitude. This places the slope problem on equal footing with the physical scatter problem as a target for calibration improvement in the MeerKAT and SKA era, a quantification not previously reported in the literature. We caution that this is a two-way decomposition, and that the noise-floor term should be read as an *upper bound* on the irreducible astrophysical scatter rather than a pure measure of it. By construction it absorbs every source of unexplained variance that is not the slope bias, including SFR and radio measurement uncertainties, photometric-redshift errors, and any residual contribution from AGN that lie below the COSMOS identification thresholds (Section 2.6). We do not attempt to separate these terms here; doing so would require propagating per-source measurement errors and an explicit contamination model, which the available data do not constrain cleanly. The $\Delta R^2 = 0.30$ floor therefore bounds, but does not isolate, the genuinely irreducible synchrotron-emissivity scatter, and the true astrophysical floor is correspondingly somewhat smaller.

Compared to the full-sample bootstrap baseline of $R_{\text{boot}}^2 \approx 0.85\text{--}0.90$ (computed using $\sigma_{\text{obs}} = 0.51$ dex over the complete $N = 7,729$ COSMOS sample), the gap confirms that the shortfall is set by the data rather than the method: radio luminosity is decoupled from SFR by an amount that no regression architecture in our suite can recover. This ceiling is unlikely to arise from methodological limitations, as it persists across linear regression, tree-based methods, kernel approaches, and neural networks. Sub-unity behaviour has previously been suggested at low radio luminosities due to cosmic-ray escape and variations in magnetic field strength (e.g., Bell 2003; Lacki et al. 2010). Our results demonstrate that this behaviour extends across environments and limits the recoverability of SFR from radio luminosity, even when flexible non-linear models are employed.

Several well-established mechanisms contribute to this decoupling. First, synchrotron emission depends on magnetic field strength and cosmic-ray confinement efficiency. In low-mass or diffuse systems, cosmic-ray escape can suppress synchrotron emission relative to infrared luminosity (Bell 2003; Lacki et al. 2010). Conversely, compact starbursts may approach calorimetric conditions in which relativistic electrons and positrons ($e^\pm$) lose most of their energy within the galaxy, enhancing radio emissivity per unit SFR (Lacki & Thompson 2010). Variations in magnetic field strength and gas density therefore introduce intrinsic scatter (Tabatabaei et al. 2013).

Second, radio and infrared emission respond on different timescales to changes in star formation history. Infrared emission traces dust heating by young massive stars, while non-thermal radio emission arises from supernova remnants associated with stars more massive than $\sim 8 M_\odot$ (Condon 1992; Arango-Toro et al. 2023). Rapidly varying or bursty star formation histories can therefore temporarily displace galaxies from the mean infrared–radio relation by $\lesssim 0.5$ dex without invoking AGN activity (Bressan et al. 2002; Delhaize et al. 2017).

Third, at higher redshift, inverse-Compton losses off the cosmic microwave background increasingly suppress synchrotron efficiency (Murphy 2009; Schleicher & Beck 2016), so the coupling between radio luminosity and SFR is expected to *weaken* rather than tighten with redshift. Although our cluster sample extends only to $z \sim 0.87$, the COSMOS sample reaches $z \sim 6.5$, and we observe no evidence of a restored tight relation at high redshift, consistent with this expectation (Magnelli et al. 2015; Delvecchio et al. 2021).

5.2.

Finally, in cluster environments, additional radio emission from relativistic electrons in the intra-cluster medium, shocks, and turbulent re-acceleration can contaminate or enhance galaxy-scale synchrotron emission (e.g., Croston et al. 2019; Cuciti et al. 2021). The systematically weaker correlations observed within R_{500} are consistent with environmental processes introducing further variance.

Taken together, these effects imply that the radio-SFR connection is fundamentally modulated by magnetic field strength, cosmic-ray transport, environment, and star formation history. A single deterministic mapping from radio luminosity to SFR is therefore physically incomplete.

5.2. Why Unity-Slope Radio-SFR Calibrations Fail

The slope of $m \simeq 0.5\text{--}0.6$, significantly shallower than the unity-slope mapping assumed by standard calibrations (Condon 1992; Bell 2003), implies luminosity-dependent biases of ~ 0.5 dex per decade in radio luminosity relative to any fixed pivot (Section 4.1). Unity-slope calibrations implicitly assume that synchrotron emission scales in direct proportion to the rate of core-collapse supernovae, and hence with recent star formation, under conditions where cosmic-ray production and energy losses are constant across the galaxy population (Voelk 1989; Lacki et al. 2010). This bias is not removed by excluding AGN, since the shallow slope persists in the non-AGN subsample, and the majority of AGN hosts and radio-excess systems lie within 3σ of the best-fitting relation, confirming they do not dominate the scatter.

To quantify the practical cost of applying a unity-slope calibration, we compare the perfect-mapping line ($m = 1$, pivot-anchored; Figure 1, orange dashed) with the empirical best-fit ($m = 0.577 \pm 0.018$, black solid). The lower panel of Figure 1 shows the systematic offset $\Delta \log_{10}(\text{SFR}) = \text{predicted} - \text{true}$ for both. At $\log L_{1.4\text{GHz}} \lesssim 22.5$, the unity-slope calibration underestimates SFR by ~ 0.4 dex; at $\log L_{1.4\text{GHz}} \gtrsim 25$, it overestimates by ~ 2 dex. Adopting the empirical best-fit slope instead ($m = 0.577$) removes this leading luminosity-dependent term, and so substantially reduces the calibration bias; however, it does not eliminate it, showing the opposite pattern at low luminosities and still diverging at high luminosities because of the off-relation radio-excess and AGN population. In terms of recoverable variance, the sub-unity slope accounts for $\Delta R^2 = 0.26$, comparable in magnitude to the irreducible noise floor ($\Delta R^2 = 0.30$). This places the calibration slope problem on equal footing with the physical scatter problem: both must be addressed for radio-based SFR inference to improve substantially beyond its current ceiling.

The progressive luminosity-cut experiment (Figure 2, Table 6) localises the slope bias to the high-luminosity end: removing the bright tail steepens the empirical slope toward unity and halves the slope-bias contribution, confirming that the sub-unity behaviour is driven by the off-relation bright population rather than being a uniform property of the star-forming sequence. Adopting the best-fit slope therefore reduces, but does not remove, the luminosity-dependent bias, because that bright population cannot be excised prospectively without independent classification (Section 2.5). The breakdown of a single global relation is therefore intrinsic rather than an artefact of extreme outliers (Lacki et al. 2010; Delhaize et al. 2017). Physically, it arises because the conditions governing synchrotron emissivity, including magnetic field strength, cosmic-ray confinement efficiency, and star formation history, vary systematically with galaxy luminosity and mass (Bell 2003; Lacki et al. 2010; Schleicher & Beck 2016), so the proportional coupling between radio luminosity and SFR assumed by unity-slope calibrations does not hold across the luminosity range probed by deep radio surveys. Any radio-only calibration that assumes a universal unity-slope mapping will necessarily introduce systematic, luminosity-dependent errors.

5.3. Radio-excess Sources and Realistic Survey Conditions

The data establish in Section 2.5 that removing radio-excess sources tightens the radio-SFR relation substantially (Table 4), but that this tightening is a logical consequence of excising sources defined by their offset from the relation, not evidence that those sources are physically distinct contaminants. The morphological characterisation (Table 3) shows that radio-excess hosts are compact, smooth, and centrally concentrated at fixed stellar mass, redshift, and SFR,

properties of passive early-type galaxies or secularly fuelled low-luminosity AGN rather than merger-triggered radio AGN, and that the independently identified AGN are retained even as the cut is applied. The relevant implication for survey science follows directly: in wide-field surveys the uniform deep X-ray and far-infrared data needed to apply an independent q_{IR} cut are unavailable (Knowles et al. 2022), so excluding radio-excess sources is neither uniformly applicable nor physically motivated and would systematically remove a real population of star-forming galaxies. Retaining them therefore yields a more conservative and realistic characterisation of radio-based SFR inference. As noted in Section 2.6, this inference rests on the COSMOS AGN census; any residual undetected-AGN component would be larger in shallower fields, which only strengthens the conclusion that radio luminosity alone is a limited SFR proxy.

5.4. Implications for MeerKAT and the SKA Era

MeerKAT and forthcoming SKA surveys will probe star-forming galaxies to sub- μJy flux densities across wide areas and cosmic time (Knowles et al. 2022). Radio continuum measurements will therefore become increasingly central to studies of galaxy evolution. However, our results indicate that radio luminosity is a limited stand-alone SFR estimator, recovering at most $\sim 40\text{--}45\%$ of the SFR variance without additional multi-wavelength context.

Even in the well-characterized COSMOS field, radio luminosity explains at most $\sim 40\text{--}45\%$ of the variance in SFR on the test set, well below the noise-limited ceiling of $R_{\text{perfect}}^2 = 0.70$. The gap between the two ($\Delta R^2 = 0.26$) is entirely attributable to the sub-unity slope, which means that even a perfect noise-limited calibration of the form assumed by Condon (1992) and Bell (2003) would still underperform by this margin. In cluster environments, the predictive reliability is further degraded. A universal unity-slope conversion from radio luminosity to SFR is therefore not supported by the data.

Importantly, the qualitative behaviour observed in the COSMOS field, where SFRs are derived via full far-infrared SED fitting that explicitly models AGN contributions (Delvecchio et al. 2017; Berta et al. 2013), is reproduced in the cluster sample despite the independent WISE-based mid-infrared SFR estimator (Cluver et al. 2017). This consistency across two independent SFR methodologies argues that the breakdown of the radio-SFR relation is not driven by the choice of infrared calibration.

To quantify the practical implications for survey science, Table 8 provides empirical 68%, 95%, and 99% confidence intervals on the predicted $\log_{10}(\text{SFR})$ as a function of radio luminosity bin, derived from the model ensemble. These intervals can be incorporated directly into uncertainty budgets for radio-based SFR catalogues.

Table 8. Lookup table for predicted $\log_{10}(\text{SFR})$ as a function of $\log_{10}(L_{1.4\text{ GHz}})$. The median predicted log SFR is reported with asymmetric uncertainties derived from the 16th and 84th percentiles of the six-model ensemble prediction distribution in each luminosity bin (i.e. median $- P_{16}$ and P_{84} $-$ median, in dex). Δ_{95} and Δ_{99} are the half-widths of the 95% and 99% empirical confidence intervals in dex, derived from the same ensemble on the COSMOS test set. N is the number of sources per bin. The highest-luminosity bin ($\log L_{1.4\text{ GHz}} > 24.42$) is dominated by radio-excess and AGN sources whose true SFR turns over (median true log SFR = 1.53, below the preceding bin) while the radio-trained prediction continues to rise; the quoted interval captures the ensemble spread rather than this luminosity-dependent bias (cf. Section 4.1, Figure 1).

$\log L_{\text{min}}$	$\log L_{\text{max}}$	$\log L_{\text{cen}}$	N	Median true log SFR	Median pred. log SFR	Δ_{95}	Δ_{99}
18.75	22.40	22.10	157	0.61	$0.61^{+0.174}_{-0.257}$	0.550	0.784
22.40	22.75	22.59	171	1.06	$0.97^{+0.112}_{-0.102}$	0.281	0.502
22.75	22.96	22.86	176	1.26	$1.16^{+0.095}_{-0.069}$	0.282	0.446
22.96	23.14	23.07	144	1.47	$1.31^{+0.096}_{-0.079}$	0.304	0.571
23.14	23.31	23.22	169	1.60	$1.42^{+0.091}_{-0.067}$	0.343	0.525
23.31	23.47	23.38	138	1.65	$1.51^{+0.105}_{-0.071}$	0.326	0.610
23.47	23.63	23.55	144	1.82	$1.62^{+0.123}_{-0.077}$	0.406	0.716

Continued on next page

Table 8. (continued)

$\log L_{\min}$	$\log L_{\max}$	$\log L_{\text{cen}}$	N	Median true log SFR	Median pred. log SFR	Δ_{95}	Δ_{99}
23.63	23.78	23.69	147	1.92	$1.74^{+0.114}_{-0.107}$	0.409	0.642
23.78	23.95	23.89	96	2.03	$1.87^{+0.121}_{-0.121}$	0.390	0.693
23.95	24.14	24.04	88	2.05	$1.94^{+0.145}_{-0.125}$	0.466	0.638
24.14	24.42	24.28	58	2.18	$2.08^{+0.147}_{-0.135}$	0.491	0.770
24.42	27.51	25.16	57	1.53	$2.21^{+0.244}_{-0.242}$	0.756	1.142

To understand which additional observables offer the greatest gains beyond 1.4 GHz luminosity alone, we perform an incremental information ladder analysis, adding features sequentially and recording the change in R^2 and the 68% confidence interval half-width at each step. Adding a second radio band (3 GHz luminosity) produces only a marginal improvement ($\Delta R^2 \lesssim 0.02$), consistent with the two luminosities carrying largely redundant spectral information under a fixed spectral-index assumption (Magnelli et al. 2015). Adding photometric redshift yields a modest but physically meaningful gain ($\Delta R^2 \approx 0.03$ – 0.05), primarily because redshift encodes the inverse-Compton loss regime and accounts for sample selection across cosmic time (Murphy 2009; Schleicher & Beck 2016). Adding stellar mass provides a similar marginal contribution. The morphological properties of radio-excess hosts documented in Table 3 (particularly the concentration, Gini coefficient, and asymmetry) encode physical conditions (gas surface density, magnetic field strength) that govern cosmic-ray calorimetric efficiency (Lacki et al. 2010; Lacki & Thompson 2010), suggesting that non-parametric morphology may represent a substantially larger information gain than any scalar radio or redshift observable. High-resolution optical imaging from Euclid, Rubin LSST, or Roman therefore represents a promising avenue for reducing radio-SFR prediction uncertainty in the SKA era.

Future radio surveys must incorporate multi-wavelength diagnostics, environmental classification, and probabilistic frameworks that explicitly account for intrinsic scatter and non-star-forming contributions. Rather than serving as a direct SFR proxy, radio continuum emission should be interpreted as one component of a multi-parameter system governed by magnetic fields, cosmic-ray physics, star formation history, and environment (Lacki et al. 2010; Murphy 2009; Schleicher & Beck 2016). Its breakdown at both high luminosities and in dense environments highlights the need for calibration strategies that are luminosity-dependent, environment-aware, and physically motivated.

6. CONCLUSIONS

We have investigated the reliability of 1.4 GHz radio continuum emission as a tracer of star formation rate (SFR) across field and cluster environments, combining deep radio observations, infrared-based SFR measurements, and supervised machine learning techniques. Our analysis spans the VLA-COSMOS field (extending to $z \sim 6.5$) and 24 galaxy clusters from the MeerKAT Galaxy Cluster Legacy Survey (MGCLS), allowing us to test the radio-SFR connection under both ideal and environmentally complex conditions. Our principal conclusions are as follows:

- Radio luminosity encodes limited information about SFR, and the information loss has two physically distinct origins.** Even when flexible non-linear models are employed, radio-only inputs explain at most ~ 40 – 45% of the variance in infrared-derived SFRs
-
- ($R^2 \lesssim 0.45$) on the held-out test set ($N = 1,546$). A perfect proportional mapping (slope unity, pivot-anchored, subject only to the observed scatter) achieves a theoretical ceiling of $R^2_{\text{perfect}} = 0.70$ on this same test set. The full variance budget decomposes as: (i) an irreducible noise floor ($\Delta R^2 = 0.30$), arising from magnetic field variation, cosmic-ray transport, and star formation history scatter that radio luminosity cannot encode; and (ii) a calibration bias ($\Delta R^2 = 0.26$), arising from the sub-unity slope ($m \simeq 0.577$) that renders any unity-slope calibration luminosity-dependent. These two contributions are of comparable magnitude, placing slope-refinement on equal footing with understanding the physical scatter. A bright-end luminosity cut steepens the empirical slope toward unity and recovers a tighter relation (Figure 2), but this does not rescue radio-only inference in practice: it removes the same off-relation population isolated by the q_{IR} cut, cannot be applied without knowing the SFR in advance, and still leaves $R^2 \simeq 0.56$, short of the $R^2_{\text{perfect}} = 0.70$ ceiling. The

full-sample bootstrap baseline ($R_{\text{boot}}^2 \approx 0.85\text{--}0.90$; $N = 7,729$; $\sigma_{\text{obs}} = 0.51$ dex) confirms that more than half of the intrinsic SFR variance is physically unrecoverable from radio luminosity alone.

4. **The radio–SFR relation has a slope significantly shallower than unity.** The best-fitting slope in log–log space is $m \simeq 0.577 \pm 0.018$, significantly shallower than the unity slope assumed by standard calibrations. A unity-slope calibration therefore introduces systematic, luminosity-dependent biases, overestimating SFRs at high radio luminosities and underestimating them at low luminosities. As shown in the lower panel of Figure 1, neither the empirical fit nor the perfect proportional mapping avoids these biases at extreme luminosities: the slope problem is intrinsic, not a consequence of sample selection. In terms of recoverable variance, the sub-unity slope accounts for $\Delta R^2 = 0.26$, comparable in magnitude to the irreducible physical scatter floor.
5. **Radio-excess sources are morphologically distinct and cannot be straightforwardly excluded.** Removing radio-excess sources does restore a tighter radio–SFR relation, but these sources cannot be unambiguously classified as AGN from radio data alone (Padovani et al. 2015). As shown in Table 3, radio-excess hosts are significantly smoother and more centrally concentrated than matched controls, consistent with early-type or passive galaxies (van der Wel et al. 2014) rather than with disturbed AGN hosts (Holt et al. 2008; O’Dea & Saikia 2021). Their exclusion from wide-field surveys is therefore not physically justified without resolved multi-wavelength classification.
6. **Dense environments degrade predictive reliability beyond the field ceiling.** Applying the field-trained models to 24 MGCLS clusters yields a distinct, environment-specific result rather than a simple repeat of the COSMOS finding: the predicted-to-observed correlation falls from $\rho \sim 0.39$ outside R_{500} to $\rho \sim 0.21$ within the cluster cores, a degradation of nearly a factor of two that the field sample cannot reveal. This is consistent with additional synchrotron contributions from intra-cluster processes and environmental modulation of cosmic-ray physics (Croston et al. 2019; Cuciti et al. 2021), and shows that radio-based SFR inference becomes progressively less reliable toward dense environments.
7. **A universal unity-slope radio–SFR calibration is not supported.** The combination of intrinsic physical scatter, a slope shallower than unity, and environmental dependence demonstrates that 1.4 GHz radio luminosity cannot serve as a precise stand-alone SFR tracer (Lacki et al. 2010; Murphy 2009; Schleicher & Beck 2016). Empirical confidence intervals on predicted SFR as a function of luminosity are provided in Table 8.
8. **Radio-excess sources are morphologically distinct and cannot be straightforwardly excluded.** Removing radio-excess sources does restore a tighter radio–SFR relation, but these sources cannot be unambiguously classified as AGN from radio data alone (Padovani et al. 2015). As Table 3 shows, at fixed stellar mass, redshift, and SFR the radio-excess hosts have significantly higher Gini coefficients ($d = +0.22$) and concentration indices ($d = +0.18$) and significantly lower asymmetries ($d = -0.14$; rising to $d = -0.87$ for the flat-spectrum sub-sample) than matched controls, an early-type, centrally concentrated, morphologically undisturbed signature consistent with passive galaxies (van der Wel et al. 2014) rather than the disturbed hosts of merger-triggered radio AGN (Holt et al. 2008; O’Dea & Saikia 2021). Their exclusion from wide-field surveys is therefore not physically justified without resolved multi-wavelength classification.

Taken together, these results indicate that 1.4 GHz radio luminosity alone is insufficient for *precise* SFR inference across luminosity and environment. Radio continuum remains a valuable tracer of galaxy evolution, but its use as a quantitative SFR estimator requires probabilistic treatment of intrinsic scatter, and, where possible, multi-wavelength or environment-aware context to mitigate luminosity-dependent and environment-dependent systematics. As MeerKAT and the SKA probe star-forming galaxies to unprecedented depths, future calibration strategies must be luminosity-dependent, environment-aware, and physically motivated.

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carried out with the Karl G. Jansky Very Large Array (VLA) of the National Radio Astronomy Observatory. This research has made use of data from the VLA-COSMOS 3GHz Large Project (Smolčić et al. 2017), and from the MeerKAT Galaxy Cluster Legacy Survey (MGCLS) (Knowles et al. 2022). We thank Kabelo Kasebonye, Shuowen Jin, and Leah Morabito for useful discussions, advice and comments. JSM is supported by the Lancaster University Faculty of Science and Technology. BDS acknowledges support through a UK Research and Innovation Future Leaders Fellowship [grant number MR/T044136/1] and its renewal [grant number MR/Z000076/1].

Facilities: VLA, MeerKAT

APPENDIX

A. ABELL 22: ILLUSTRATIVE CLUSTER EXAMPLE

Figure 7 shows the predicted-to-observed flux density ratios and Spearman correlations for Abell 22, an illustrative example of the behaviour observed across the full MGCLS cluster sample. Results for all sources, sources within R_{500} , and sources outside R_{500} are shown. The pattern of weakening correlation within the cluster core is characteristic of the full sample (Figure 5).

B. DATA AVAILABILITY

The raw data used in this study are all publicly available at <https://heasarc.gsfc.nasa.gov/W3Browse/all/vlacosxoid.html>, <http://mgcls.sarao.ac.za>, and <https://vizier.cds.unistra.fr/viz-bin/VizieR?-source=II/328>. Scripts underlying this study are available at <https://github.com/JasonMakechemu/Radio-Luminosity-as-a-star-formation-tracer>.

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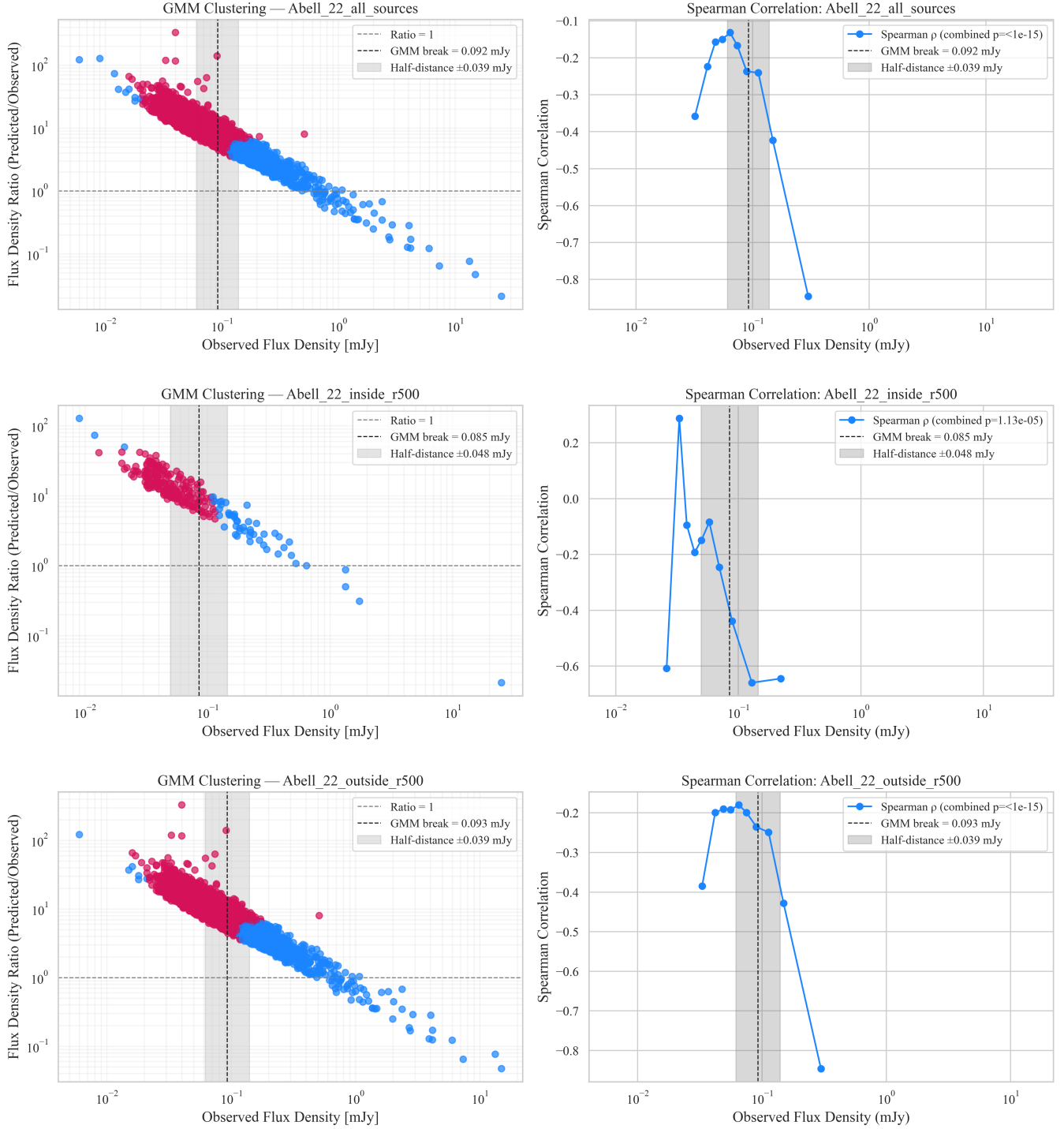


Figure 7. Predicted-to-observed flux density ratios (mJy/mJy) and Spearman correlations for Abell 22 for all sources (top), sources within R_{500} (middle), and sources outside R_{500} (bottom). Similar behaviour is observed across the full cluster sample (Figure 5).

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